

Project: Computer Science

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Topic : How SMEs Use SEO to Compete in Digital
Markets Importance of SEO in Digital Marketing for
Small Businesses

Declaration

I hereby declare that this thesis entitled “The Role of Search Engine Optimization (SEO) in Enhancing the Competitiveness of Small and Medium-Sized Enterprises (SMEs) in Digital Markets” is my own original work and has been completed in accordance with the academic regulations of the university.

This thesis has not been submitted previously, either in whole or in part, for the award of any other degree or qualification at any other institution. All sources of information used in this research have been appropriately acknowledged and cited in the reference list.

I confirm that this work is the result of my independent research and that any assistance received has been properly acknowledged.

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Abstract

The intensive growth of the digital economy has greatly shaped the competitive landscape of small and medium-sized enterprises (SMEs), and it is time to value digital marketing proficiencies in the process of ensuring business success. Search engine optimization (SEO) is one of these abilities that have turned out to be cost-efficient and strategically significant to help firms to enhance their online presence, find new clients, and enhance their competitive niche in the digital markets. This research explores the way of using SEO as a digital marketing tool by SMEs and its effects on the competitiveness and performance of firms.

The study is based on the Technology Acceptance Model (TAM) and the digital capability approach. The suggested relations between the adoption of SEO, digital marketing capability, SME competitiveness, and firm performance were tested through a positivist research philosophy and a deductive research approach. Structured online survey was used to collect primary data that was collected to SME owners and marketing managers. The 150 valid responses were subjected to quantitative statistical tools such as descriptive statistics and correlation analysis.

The results indicate that the use of SEO has a positive impact on the online presence of SMEs, their ability to acquire customers, credibility of their brands, and the competitiveness on the market overall. The rate of adoption of SEO within firms however differs between firms based on their level of digital maturity, availability of resources, and knowledge of management. The findings are also that the most widely applied features in SEO are content development and optimization of the key words and that other advanced features like technical SEO and creation of back links are not regularly applied because of the financial and knowledge limitations. Moreover, the researcher affirms that the perceived usefulness and the perceived ease of use are both central variables that determine the adoption of SEO, which underscores why the Technology Acceptance Model can be applicable in the context of digital marketing of SMEs.

The study is also important to literature by defining SEO as a strategic digital competency that promotes SME competitiveness and merging technology adoption theory with the digital transformation and resource-based views. The results also have practical implications to SME managers and policymakers who would like to facilitate digitalization and innovation in small businesses.

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List of Abbreviations

AI – Artificial Intelligence
B2B – Business-to-Business
B2C – Business-to-Consumer
CEO – Chief Executive Officer
CRM – Customer Relationship Management
DOI – Digital Object Identifier
EU – European Union
GDP – Gross Domestic Product

ICT – Information and Communication Technology

IT – Information Technology

KPI – Key Performance Indicator

MIS – Management Information Systems

ROI – Return on Investment

SEM – Search Engine Marketing

SEO – Search Engine Optimization

SERP – Search Engine Results Page

SME – Small and Medium-Sized Enterprise

SMM – Social Media Marketing

SPSS – Statistical Package for the Social Sciences

TAM – Technology Acceptance Model

TOE – Technology–Organization–Environment

URL – Uniform Resource Locator

UX – User Experience

WWW – World Wide Web

1. Introduction

1.1 Background and Research Context

The explosion of digital technologies has greatly changed how business is conducted, communicated, and competed in modern markets. The introduction of the digital space, data-oriented marketing, and search engine-based visibility has transformed the definition of the competitive advantage to rely on the digital capability and online presence rather than the traditional physical assets (Verhoff et al., 2021; Vial, 2019). Since most consumers are relying more on online information to guide them in the buying process, companies that are interested in staying competitive in the current business contexts should embrace the use of digital marketing strategies. Digital marketing allows organizations to communicate with a larger number of prospective customers, address them more personally, and measure the marketing performance based on quantifiable metrics and analytical tools (Kannan and Li, 2021; Li et al, 2021).

In this online world, small and medium-size businesses (SMEs) are the most important aspects in economic growth, innovation, and job creation. The SMEs constitute a significant percentage of all businesses across the globe and they play an important role in the economic development of the country. Nonetheless, SMEs are as a rule disadvantaged in terms of resources (financial resources, lack of technical knowledge and access to sophisticated marketing technologies) compared to large corporations (Kraus et al., 2021; Müller et al., 2021). These constraints render the SMEs to compete effectively in highly digitalized and data intensive markets difficult. In turn, the issue of digital transformation is not only an opportunity but also a need of the SMEs to be into the long-term sustainability and competitiveness (Matarazzo et al., 2021; Eggers et al., 2020).

The digital transformation enables SMEs to utilize digital technologies to become more efficient in their operations and to have better customer relations, as well as increase their market reach. The search engine optimization (SEO) is one of the few digital marketing tools that have assisted SMEs to become more visible online and reach their potential clients more affordably. SEO is the act of increasing the ranking of a webpage in search engine results pages (SERPs) by enhancing the content, structure, and technical performance of the webpage. The greater the rankings, the more likely that the potential customers will find the products or services of a firm when searching on the internet.

SEO is an especially appealing opportunity to SMEs with modest marketing budgets since it does not rely on paying to promote their services but on the organic visibility on search engines. By optimizing keywords, creating high-quality content, and improving the technical aspects of the site, firms will be able to generate more traffic to their websites and raise their brand awareness without having to spend significant sums of money on advertising (Berman and Katona, 2019; Ahmad et al., 2024). With search engines like Google becoming the main point of entry to the information on the Internet, the ability to be ranked high on the search page has become a crucial success factor in digital marketing.

Recent works suggest that digital marketing skills such as SEO, social media presence, and data-based marketing have the potential to enhance the performance of firms, their interactions with customers, and innovation capacity to a high degree (Scuotto et al., 2021; Sun et al., 2022; Kumar et al., 2023). Through these digital tools, the SMEs will be able to compete with other larger firms in terms of increasing visibility in the online markets, as well as accessing new customer segments. Moreover, the digital marketing will allow SMEs to gain access to international markets with less hassle, which will allow internationalization and market expansion where smaller businesses could previously struggle to accomplish (Zhou et al., 2020).

Even with these possible advantages, the perception of digital marketing technologies among the SMEs is not even. Most small businesses have difficulties in adopting more sophisticated digital strategies because of organizational, technological, and environmental constraints. SMEs do not exploit the opportunities of digital marketing to a full extent due to limited managerial knowledge, absence of technical expertise, and a lack of understanding of whether the investment will pay off (Chatterjee and Kar, 2020; Alford and Page, 2019). Consequently, some of the SMEs have been able to incorporate digital marketing in their business strategies, whereas some are still in the initial stages of digitization.

1.2 Problem Statement

Even though digital marketing and digital transformation have gained more interest in scholarly studies, a significant part of the literature available centers around large institutions with a considerable amount of technological and financial capabilities. Big companies usually have their own marketing departments, powerful analytics, and budgetary resources to execute complicated digital marketing plan. By contrast, SMEs often do not have the resources or knowledge to

implement and introduce digital marketing technologies in an effective manner (Kraus et al., 2021; Cichosz et al., 2020).

The given gap poses a considerable contradiction between the perceived benefits of digital marketing and the real capacity of the SMEs to apply these technologies. Although online visibility and competitive rates in the market can be increased through digital methods like SEO, most of the SMEs lack the capacity to adopt them strategically because of knowledge and resource availability.

SEO has been known to be a pivotal power of Internet presence and clientele. Nonetheless, successful execution of SEO presupposes constant optimization, planning, and analytical skills that most SMEs can hardly have (Ahmad et al., 2024; Berman and Katona, 2019). Additionally, even though SEO is a common topic of research in digital marketing, very little empirical evidence exists to investigate the connection between the adoption of SEO and the competitiveness of SMEs.

1.3 Research Aim and Objectives

The main aim of this study is:

To examine how SMEs use search engine optimization (SEO) as a digital marketing strategy to enhance their competitiveness and performance in digital markets.

To achieve this aim, the study pursues the following objectives:

1. To analyse the role of digital marketing in the competitiveness of SMEs.
2. To examine the importance of SEO in improving online visibility and customer acquisition.
3. To identify the key SEO practices adopted by SMEs.
4. To investigate the challenges SMEs face in implementing SEO strategies.
5. To evaluate the impact of SEO on SME performance and market competitiveness.

1.4 Research Questions

This study addresses the following research questions:

1. How do SMEs use SEO as part of their digital marketing strategy?

2. What are the key benefits of SEO for SME competitiveness and performance?
3. What challenges do SMEs face in implementing SEO?
4. How does SEO contribute to SMEs' ability to compete in digital markets?

1.5 Academic and Practical Relevance

This research has several contributions to the academic research. First, it broadens the current literature on digital marketing and the performance of SME by exploring the concept of SEO as a strategic capability and not just a technical marketing device. It is not the first study to discuss the significance of digital capabilities in enhancing the performance of firms (Khin and Ho, 2019; Wamba et al., 2020), but comparatively little research has researched how particular digital marketing techniques like SEO affect the competitiveness of SMEs.

Second, the research is valuable to the literature review on the concept of digital transformation because it focuses on how SMEs simultaneously adopt digital marketing strategies as part of their business models. Past studies indicate that digital technologies can help companies enhance operational efficiency, develop new value propositions, and build better customer relationships (Rachinger et al., 2019; Reim et al., 2020). This research gives more comprehensive information on how digital marketing abilities facilitate competitiveness of SMEs by undertaking the analysis of adopting SEO.

In practice, the research is helpful to the managers of SMEs and practitioners of digital marketing. The research provides insights on how SMEs can use digital marketing tools to make use of their scarce resources and have sustainable competitive advantage by identifying effective SEO practices and the challenges that come with their implementation (Saura et al., 2021; Gavrila and de Lucas Ancillo, 2021).

1.6 Structure of the Thesis

In this thesis, six chapters have been subdivided.

Chapter 1 -Introduction: This chapter includes the research background, problem statement, research aim and objectives, research questions, and the relevance of the research both academically and practically.

Chapter 2- Theoretical Framework and Literature Review summarizes the prior literature on SMEs in the digital economy, digital marketing and firm performance, the concept and evolution of SEO, SEO techniques and issues, the Technology Acceptance Model (TAM), digital marketing capability, SEO and customer acquisition, and the conceptual framework with research hypotheses.

Chapter 3 -Methodology details the research design, which entails the research philosophy, deductive approach, survey approach, sampling method, data collection strategy, questionnaire design, pilot study, data screening strategy, and statistical analysis strategy, to be employed in the study.

Chapter 4, Empirical Results, is a presentation of the survey analysis results, such as sample characteristics, level of SEO adoption, and the SEO techniques employed by SMEs, the challenges in adoption, the reliability testing, the regression analysis, and graphical analysis of the key variables.

Chapter 5 -Discussion interprets the results with respect to the existing literature and talks about the implications of the results in terms of theoretical implications, managerial implications, digital strategy implications, and policy implications.

Chapter 6 -Conclusion summarizes the primary research findings and provides the study contributions, limitations, future research directions, and policy implications.

2. Theoretical Framework and Literature Review

2.1 SMEs in the Digital Economy

SMEs are very important in the contemporary economies since they provide employment, innovation, and economic development. SMEs in most countries have numerous businesses and a significant proportion of the total employment and activity. The SMEs have limited financial resources, technological resources, and managerial resources, though they are important to the economy. The above restrictions complicate the adjustment of SMEs to the rapidly changing market environment, especially when it comes to digital transformation (Kraus et al., 2021; Müller et al., 2021).

The advent of digital economy has radically transformed the business competitive environment. The online platform, cloud computing, and online digital marketing tools are digital technologies that have altered the old business model and reformulated the way businesses generate and deliver value. In the digital economy, competitive advantage is becoming based more on the capacity of firms to create digital capabilities, capitalize on data-driven choices, enhance good visibility on the internet (Verhoef et al., 2021; Vial, 2019). This has made the implementation of digital technologies to be a necessity of SMEs who want to be competitive in the contemporary markets.

There are a few opportunities of digitalization to SMEs. Embracing digital technologies, the firms can increase the market, enhance communication with the customers, and optimize the work. Specifically, digital platforms allow SMEs to reach global markets and compete with bigger organizations by targeting a niche and providing specialized services or goods. SMEs are usually quite flexible and agile and thus can react swiftly to customer needs as well as technological change (Cenamor et al., 2019; Zhou et al., 2020). Consequently, digital transformation has the potential to help SMEs break down market barriers in the traditional world and engage in the global economy more.

Nonetheless, despite these opportunities, some SMEs encounter considerable problems with embracing the digital technologies. Low financial capabilities, deficiency of technical skills and digital capabilities usually create delays in the digital transformation process. A lot of small companies cannot access a high-level of technological background and specific people needed to develop complex digital strategies (Cichosz et al., 2020; Eggers et al., 2020). These obstacles bring

in differences in digital adoption among SMEs as some of the firms have managed to integrate digital technologies into their lives whereas some are still at the initial phase of digitalization.

The other noteworthy element of the digital economy is the increased significance of digital marketing capabilities. The digital marketing tools enable the firms to access their customers more easily, study consumer behavior and gauge the efficiency of marketing strategies in real time. The emergence of data analytics and online marketing campaigns allows companies to create tailored marketing campaigns and enhance consumer interaction (Sun et al., 2022; Trainor et al., 2020).

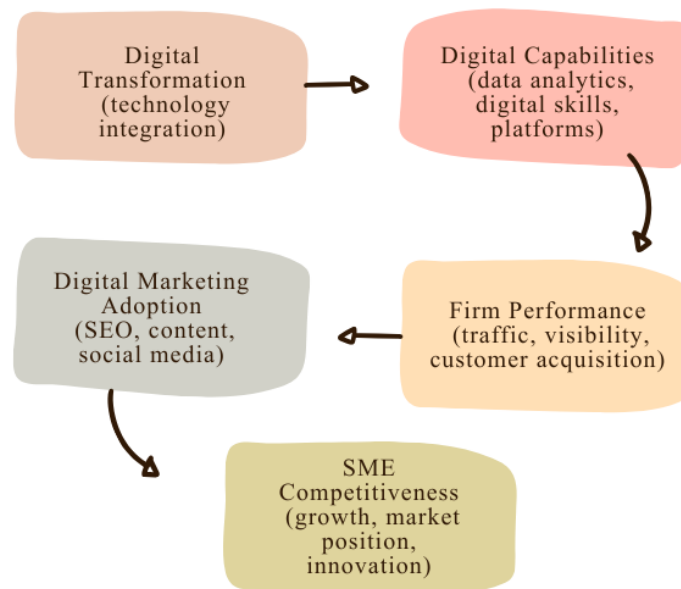


Figure 1: Digital transformation and SME competitiveness model.

Source: Adapted from Verhoef et al. (2021), Vial (2019), and Kraus et al. (2021).

Figure 1 shows the correlation between digital transformation, development of digital capability, and competitiveness of SME. Digital transformation allows the firms to acquire digital capabilities, which facilitate effective execution of digital marketing strategies. These features enhance the performance of the firm and eventually contribute to competitiveness in the digital markets (Verhoff et al., 2021; Vial, 2019; Kraus et al., 2021).

2.2 Digital Marketing and Firm Performance

The modern business world has seen digital marketing as one of the greatest contributors to the performance of firms. As the internet technologies and online platforms rapidly develop, the number of organizations that are turning to the use of digital channels to involve the customers, advertise their products and services, and amplify the brand awareness increases. Digital marketing will allow the firms to access bulk populations of people via different online platforms like search engines, social media platforms, websites, and email marketing among others. Through these tools, firms may carry out targeted and quantifiable marketing campaigns, which can be tracked and optimized on the fly (Kannan and Li, 2021; Li et al., 2021).

Digital marketing is particularly a great opportunity to compete in more digitalized markets in the case of small and medium-sized enterprises (SMEs). In contrast to the traditional marketing techniques like television ads, print publications, and payments on the massive promotion campaign, digital marketing can contribute to less financial investments, and the companies can differentiate categories of customers more efficiently. This is the ease and flexibility that makes digital marketing to be especially appropriate to SMEs that have limited marketing resources and funds (Alford and Page, 2019). Using digital tools, SMEs can gain more awareness on the Internet, gain new customers, and reinforce their presence in the market without imperative spending on marketing.

An emerging base of empirical studies indicates that adoption of digital marketing has high positive influence on the performance of the SME. The digital marketing activities enable the firms to increase their market share, improve customer interaction, and optimize their sales. To illustrate, Sharabati et al. (2024) reveal that digital marketing is a key factor in improving the performance of firms by making the brand more recognized, ensuring that the relationships are enhanced, and that the expansion of the market can be made easier. Their research reports that SMEs which incorporate the aspect of digital marketing in their business models have higher chances of enjoying a better financial performance and competitive advantage.

Moreover, digital marketing will allow SMEs to communicate with their clients more efficiently and collect important information regarding consumer behavior. Online channels bring about huge volumes of data that an organization can interpret to determine customer preferences, buying habits, and the degree of engagement. Such insights enable companies to create tailored marketing

campaigns and enhance customer experiences that can eventually result in increased customer satisfaction and retention (Sun et al., 2022; Trainor et al., 2020). Consequently, digital marketing does not only help when it comes to customer acquisition, but also enhances the long-term relationships between the firms and their customers.

Innovation and value creation is also another significant merit of digital marketing. According to Scuotto et al. (2021), digital marketing capabilities should be used to support knowledge acquisition and sharing of information in organizations. As a result of interactions with customers, as well as by analyzing the digital data, firms can discover new trends and create innovative products or services that can satisfy the needs of customers better. This capability to create and use market knowledge can contribute to the strategic decision-making of firms and is able to lead to a long-term competitive advantage.

On the same note, Kumar et al. (2023) point out that companies which are highly equipped in digital marketing are more likely to record high performance levels. Digital marketing also allows organizations to be better positioned in understanding the market dynamics, tracking the activities of the competitors as well as responding to the changes in customer preferences in a faster manner. Consequently, companies that have a superior digital marketing framework will be able to be more strategic and adjust their marketing approaches.

The advantages of digital marketing on the performance of firms, however, are not automatic. The success of digital marketing strategies is determined by whether organizations can work out complementary capabilities that facilitate digital transformation. These abilities are data analytics proficiencies, digital leadership, organizational flexibility, and the technological infrastructure (Maroufkhani et al., 2022; Quinton et al., 2021).

2.3 Concept and Evolution of SEO

Search engine optimization (SEO) is an inherent part of the contemporary digital marketing strategy. With the growing dependence of business on internet-based strategies to connect with customers, the capacity to be ranked high in the results of search engine has been a prerequisite to securing potential customer interest and the rise of awareness about the brand. SEO is the method of streamlining the contents, hierarchy, and technological aspects of websites to enhance their positions on the list of search engines results pages (SERPs) and the prevalence of organic traffic

to websites (Berman and Katona, 2019). By improving the visibility of a site in search engines like Google, companies can capture the attention of potential consumers who are actively searching in search engines to get a particular product or service.

The role of SEO has increased tremendously with the growth of the internet and the rising normalcy of consumers using the internet to seek information and make buying decisions that require use of the search engines. Search engines are now the main point where people get to learn about the contents, products, and services in the internet. As a result, organizations that are ranked higher on the search list have a higher chance to capture people with their websites, generate leads, and turn such leads into customers. In the case of businesses that are trading online in competitive markets, good practice of search engine optimization can thus provide a rich source of competitive advantage.

The SEO practices were relatively less complex historically and were centered on the optimization of keywords. With the initial phases of search engine evolution, web pages were able to rank high by adding the keywords of relevance as many times as possible in the webpage content and metadata. Such a strategy is commonly known as “the keyword stuffing strategy; it was a popular method of SEO in the early days since the search engine algorithms mainly used the frequency of keywords to establish the relevance of webpages (Berman and Katona, 2019). Nevertheless, with development of search engines, the changes in the algorithms brought more advanced ranking parameters to refine the quality of the search result and avoid manipulating.

SEO has also become more complex and strategic as it is nowadays, occupying several dimensions of optimizing websites. The current methods of SEO are not limited to their keywords but also the quality of content, user experience, site performance, and its technical optimization. The current search engines consider far more variables in ranking the websites and these factors include content relevance, the authority of the site, user interaction factors, mobile compatibility, and page loading speed. With these developments, organizations have been encouraged to be more holistic in their approach to SEO, which focuses on the development of valuable content and positive user experiences.

The other major advancement in the field of SEO is the introduction of artificial intelligence and data analytics to search engines. The new generation of search engines takes advantage of the improved machine learning algorithms to comprehend the intent of the user and provide more

relevance to the search. This means that companies are forced to concentrate in creating high quality, informative and user-friendly content as opposed to only playing with technical search algorithm manipulation. This change has moved SEO out of a technical practice and more of a strategic marketing practice that involves the use of content marketing, web design and data-driven decision making.

SEO is one of the most useful on-line marketing tools to SMEs. The organic search result is unlike the paid advertising strategies, which are expensive and always need financial input, as long-term visibility and sustainable growth in traffic can be achieved using organic search results. This feature renders SEO especially appealing to SMEs that have small marketing budgets. Through the adoption of efficient SEO plans, the small companies will be able to enhance their presence in the online market, connect with potential clients more efficiently, and compete with bigger organizations in the online market (Ahmad et al., 2024).

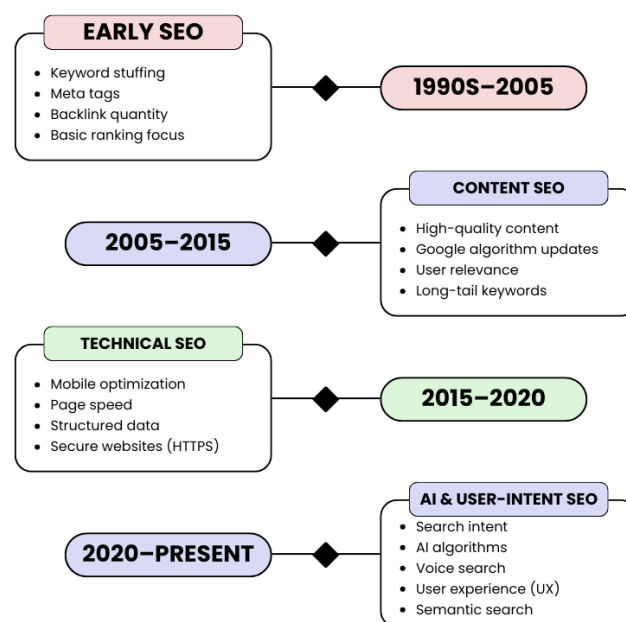


Figure 2: Evolution of search engine optimization

Source: Adapted from Berman and Katona (2019), Järvinen and Taiminen (2020), and Lewandowski (2021).

As Figure 2 shows, the key milestones in the development of SEO include the initial search tools involving a single keyword and the modern methods, which consider artificial intelligence,

semantic search, and intent. Gradually, the development of search engines algorithms has changed the emphasis to less technical manipulation up to the development of quality content, better user experience, and relevance to the search query of the user (Berman and Katona, 2019; Jarvinen and Taiminen, 2020).

2.4 SEO as a Source of Competitive Advantage

The concept of search engine optimization (SEO) can be considered both as a technical marketing instrument and strategic competency that helps firms to gain sustainable competitive advantage in digital markets, especially small and medium-sized business organizations (SMEs). With the continuous growth of the digital competition, businesses are growing more dependent on the visibility and search engine ranking as the primary way of attracting customers and reinforcing their market presence. Companies that successfully apply SEO techniques will be able to increase their visibility in search engines, reach the desired traffic, and increase customer interactions. As a result, SEO has emerged as an important way of competition among organizations in the digital economy (Berman and Katona, 2019; Ahmad et al., 2024).

Empirical studies on the connection between SEO and marketing performance indicators have shown a positive presence of evidence. Research indicates that the good practice of SEO leads to an increase in search rank, traffic to a web site and the amount of conversion. When firms advance the ranking of a site on the search results pages (SERPs) of search engines, they stand a chance of more people visiting their websites whenever they are searching the websites in case, they need their services or products. Increased organic traffic normally leads to customer acquisition and sales performance. Because of this, SEO has emerged as a valuable component of the overall efficacy of digital marketing strategies.

The resource-based view (RBV) of the firm can be used to explain the role of SEO in the creation of competitive advantage in the strategic management of the firm. According to the RBV, organizations end up attaining sustainable competitive advantage when they have valuable, rare, inimitable, and non-substitutable resources (Khin and Ho, 2019). Regarding the digital marketing, the assets related to SEO, including quality content on the site, authoritative links, streamlined web design, and data-driven optimization procedures can be viewed as strategic assets that help a company to rise in the competitive environment. The competitors can also find it hard to duplicate

these digital assets fast particularly when they have been built over time through unremitting optimization and amassing of knowledge.

One of the most strategic resources that are related to SEO is high-quality content. The search engines are placing more emphasis on information-based, relevant, and user-centered materials in ranking websites. Companies that engage in the creation of valuable content regularly are more probable to achieve a certain degree of authority in their business and earn the trust of online users. This confidence leads to better brand credibility and consumer loyalty which are the key factors in long term success in digital markets. In addition, official references of credible websites may contribute to the credibility of the domain and the position in the search engine, as well as the competitive role of a company may be increased.

The second benefit of SEO is that it offers a long-term presence as opposed to the paid digital advertisement. Though paid advertisement campaigns can achieve instant outcomes, they need incessant financial resources to sustain the appearance. SEO is however based on organic search rankings, and this can go on to drive traffic at a prolonged duration even after a site has attained good ranks. This is a long-term impact of SEO; hence it is very useful to SMEs that have limited marketing budgets. Through investing in SEO, SMEs have the capacity of developing sustainable digital properties that will keep the potential customers contacting.

SEO also allows the SMEs to compete well with the larger firms, through the niche markets and long-tail keywords. Huge companies usually take over very competitive keywords because they have huge marketing budgets and they are a recognized brand. Nevertheless, SMEs can target more targeted search terms which are representing niche customer demands or specialized products. Long-tail keywords are usually less searched but with better conversion possibilities as they reflect more user intent. When SMEs strategically focus on these keywords, it is possible to find highly relevant audiences and have good positions in special areas of the market.

Besides enhancing the visibility of the market, SEO will help in enhancing customer engagement and experience. The algorithms of modern search engines are built in such a way that it favors websites with quick loading time, mobile friendliness, and easy navigation. Consequently, companies that invest in SEO tend to enhance the usefulness and functionality of their websites. These additions increase customer satisfaction as well as chances of repeat visits which further bolster the competitive advantage.

2.5 SEO Techniques Used by SMEs

The process of search engine optimization (SEO) can be done by using a mix of on page, off-page and technical optimization strategies which together enhance the search engine ranking of a given site. These tricks are used to improve the relevance, authority, and usability of the web pages to make the web pages indexed and ranked better in the search engines. To the small and medium-sized enterprises (SMEs), proper use of the technique of SEO can greatly boost online presence and customer growth to compete competently in the digital markets.

On-page optimization is one of the most popular SEO methods and it aims at enhancing the content and structure of single webpages. On-page SEO entails optimization of keywords, titles, headings, meta descriptions, and internal links so that the web pages are pertinent to search query made by the users of the web pages. The optimization of the key words is important to make the search engines know what and why the web content is relevant. SMEs usually undertake keyword research to know the words that the potential customers are likely to use when searching products or services. Through a strategic placement of these keywords in the content of websites, SMEs stand a high chance of their webpages being listed on the web pages of search engine results (SERPs).

Content optimization is another significant element of on-page SEO. Good and informative content serves the purpose of making businesses valuable to users besides enhancing the ranking of the search engines. The content optimization involves preparation of blog posts, articles, product description, and information pages that respond to the needs and interests of target audiences. Ahmad et al. (2024) state that content creation and key optimization are often the main SEO practices that SMEs implement because they need to improve brand positioning and gain online presence. The good content developed by them does not only generate traffic on the search engine but also enhances the customer interaction as the information they develop is useful and helps in the decision-making process of buying them.

Besides the on-page optimization, off page optimization methods are crucial to build authority and credibility of the websites. The main components of off-page SEO are to acquire backlinks by other websites. Backlinks prove to be the indicators of trust and authority since search engines view them as the sign of approving content of the websites. The websites to which the reputable and authoritative sources give the links tend to be taken more seriously as a rule and, thus, are

placed higher in the search results. In the case of SMEs, the creation of partnerships, guesting, and online community are the most typical methods to gain backlinks and domain power.

Off-page SEO is also dependent on social indications and online reputation. Social media interaction has the potential to put more digital content in the view of the targeted audience and generate traffic to websites. Although the activity of social media does not directly affect the ranking of search engines, it may be used in indirect support of the work of the search engine optimization by spreading the content better and creating additional backlinks.

Technical optimization is another important aspect of SEO that aims at enhancing the performance and access of the websites. Technical SEO makes sure that the search engines can crawl and index the contents of the websites. Other important factors of technical SEO would be to enhance the speed of loading the website, improve its mobile responsiveness, improve the architecture of the site, and use a structured data structure. Search engines have been making search results based on mobile-friendly websites in case of increasing searches done with mobile devices. Those SMEs which invest in the mobile optimization and the web-pages which load quickly will thus have more chances to have a high position on search engine results.

Technical SEO will also help to increase the user experience which now is a significant ranking criterion by search engines. Websites that are quick to load, offer easy navigation and have clear information are more likely to have low bounces and high levels of engagement. All these are indicators to search engines that the site has useful and relevant information to the users.

2.6 Challenges in SEO Adoption

Despite the high advantages that search engine optimization (SEO) has on enhancing online presence and acquisition of customers, most of the small and medium-sized enterprises (SMEs) face numerous challenges in adopting and implementing the SEO strategies. These are the challenges, which usually restrict the possibilities of the SME to use SEO as a powerful digital marketing instrument to the full extent. The barriers to adoption of SEO are often limited financial resources, the absence of technical expertise, digital skills, as well as organizational constraints that do not allow strategic digital transformation (Sharabati et al., 2024; Kraus et al., 2021).

Resource limitation is one of the major challenges experienced by SMEs. SMEs have a limited ability to invest in digital marketing technologies and professional SEO services due to the limited

budgets and the number of their employees compared to large organizations. Effective optimization of a website may need ongoing efforts into the development of the sites, page creation, the tools of analytics, and special knowledge. A lot of SMEs simply cannot afford to retain the services of specific digital marketing experts or outsource the process of SEO to specific agencies. This has led to the construction of SEO on SMEs based on elementary optimization measures instead of intensive online marketing plans.

The lack of technical knowledge and digital skills is another significant challenge. SEO is a very complicated and constantly changing sphere that involves the knowledge of search engine algorithms, research of keywords, the structure of the webpage and data analytics. Most of SME managers and employees lack enough knowledge on these technicalities and therefore, effective SEO strategies can hardly be prepared and applied. Kraus et al. (2021) state that the lack of digital competencies among the employees of SME is one of the biggest obstacles to digital transformation. Lack of proper training and knowledge can cause the businesses to fail to use highly effective SEO practices like technical optimization, structured data application, or performance analytics.

Besides resource and knowledge limitations SMEs also have challenges associated with dynamism of search engine algorithms. Search engines often upgrade their ranking systems to enhance quality of search results and avoid manipulations. These changes can greatly influence the ranking of websites and they must keep on changing their SEO strategies. SMEs do not have enough time and knowledge to observe and track the changes in the algorithms and introduce alterations to optimize their practices. As a result, it might be difficult to sustain a proper performance of SEO in small companies, which lack digital marketing teams.

The other challenge is that it is not easy to gauge the payback or the return on investment (ROI) of the SEO activities. The performance outcomes of the paid advertising campaigns can be witnessed instantly, whereas the outcomes of the SEO are usually realized over time. The implementation of SEO strategies is usually a long-term process that does not bear any visible results in terms of improvement of search positions and traffic. This lagging effect poses a challenge in determining the success of investments in SEO by the SMEs and can demoralize managers to invest in SEO programs.

Another factor that is critical to the adoption of SEO is organizational factors. The implementation of digital marketing strategies in SMEs may be slowed down by resistance to change, inadequacy strategic orientation, and managerial support (Quinton et al., 2021). In most instances the digital tools are adopted in disjointed fashion without involving them in a consistent marketing plan by the SMEs. SEO efforts will not be consistent and will not yield significant performance results when these digital efforts are not coordinated with the larger business goals.

2.7 Technology Acceptance Model (TAM)

Technology Acceptance Model (TAM) is one of the best theoretical frameworks that can be used to explain how new technologies are adopted within an organization. TAM, which was initially formulated by Davis (1989) describes the process through which users accept and adopt new technologies depending on two main determinants that include the perceived usefulness and the perceived ease of use. Perceived usefulness is the degree to which people believe that the utilization of a specific technology will increase their job performance whereas the perceived ease of use is the degree to which the technology is perceived to be easy to learn and use. The two factors affect the attitude of the users towards the technology which in turn affect the behavioral intentions of the user to adopt and use the technology.

Various areas of information systems, digital marketing, and technology adoption studies have been extensively used over time with the application of TAM. The model is a basic and at the same time an effective model in analyzing decisions made by individuals and organizations to use new technological tools or not. TAM has been commonly employed to study the adoption of various technologies in the digital transformation, including online shops, cloud computing, mobile apps, and online promotional tools (Alalwan, 2020; Chatterjee and Kar, 2020).

TAM can be used to explain why some firms use digital marketing technologies and others are just not convinced to use them when applied to small and middle-sized enterprises (SMEs). Limited funds and limited technical knowledge coupled with perceived threat of technological investment tend to put SMEs in a dilemma whenever it comes to assessing new technology. As per to TAM, SMEs will embrace the use of digital marketing tools when the decision-makers perceive that they will serve as important in enhancing business performance and when the technologies are believed to be simple to adopt and use.

TAM, in this research paper, is offering a theoretical perspective on the adoption of search engine optimization (SEO) as one of the digital marketing strategies. Those managers of SMEs who view SEO as an effective tool of growing the traffic to their websites, gaining new clients, and enhancing their visibility have higher chances to include SEO as the part of their marketing campaigns. Equally, in situations where SEO tools and methods are perceived as somewhat easy to learn and apply, SMEs will embrace them as a part of their digital marketing process (Alalwan, 2020).

In addition, TAM puts significance on managerial perceptions and organizational readiness to influence technology adoption decision. SME managers that are aware of the strategic importance of SEO and have adequate digital awareness have higher chances of investing in SEO initiatives. On the other hand, the adoption levels can be low when the managers view SEO as complicated or have low confidence about the results that it will bring. This view is consistent with the existing literature that suggests the high impact of the attitude toward usefulness and ease of use on the use of digital marketing technologies among SMEs (Chatterjee and Kar, 2020).

2.8 Research Gap

Though the focus of the research has already been on digital marketing, digital transformation, and performance of small and medium-sized enterprises in the past, the strategic importance of search engine optimization (SEO) in increasing the competitiveness of small and medium-sized businesses has received comparatively little attention. However, most of the available literature is general about digital marketing tools, including social media marketing, e-commerce tools, or general development of digital capability, and the effect of SEO as a specific digital marketing approach is under-researched (Scuotto et al., 2021; Kumar et al., 2023).

Moreover, most of the studies focusing on the impact of digital transformation on SMEs focus more on the technology adoption and innovation outcomes without directly studying the effects of digital marketing practices on the competitive advantage and performance of a firm (Kraus et al., 2021; Maroufkhani et al., 2022). Consequently, the connection between the use of SEO, digital marketing competence, and competitiveness in SMEs has not been adequately studied in empirical studies.

The other research gap identified in the literature is associated with the incorporation of the technology adoption theories into the digital marketing strategy study, including the Technology

Acceptance Model (TAM). Although TAM has been a popular framework in the explanation of the adoption of information technologies, limited literature has adopted the framework to analyze the adoption of SEO practices in SMEs (Alalwan, 2020).

Thus, the paper fills these gaps with a discussion on the role of the adoption of SEO by SMEs in the context of their digital marketing and its effects on competitiveness and performance in the digital market.

2.9 Conceptual Framework

According to the current body of knowledge on digital marketing, technology adoption, and SME competitiveness, this paper will outline a conceptual framework that will explain the role of search engine optimization (SEO) in achieving competitiveness and performance of small and medium-sized enterprises (SMEs). The framework combines the knowledge of digital marketing research and technology adoption theories to demonstrate the connection between SEO activities, digital marketing capability, and firm performance.

The theoretical framework presupposes that the implementation of SEO practices is a highly important factor in promoting the digital marketing opportunities of SMEs. Online visibility and generation of organic traffic to business websites through SEO methods, including optimization of the key word, development of the content, creation of back links, and optimization of the technical websites. The enhancements reinforce the digital presence of firms and facilitate their ability to interact better with customers in the digital markets (Ahmad et al., 2024; Sun et al., 2022).

Subsequently, better digital marketing opportunities can be translated into greater competitiveness and performance of SMEs. The successful application of SEO strategies will put firms in a better position of recording increased traffic in their websites, improved customer acquisition, and brand recognition. These results will assist in building competitive advantages in the realm of digital markets and lead to the enhanced performance of business (Kumar et al., 2023; Scuotto et al., 2021).

The framework also takes into consideration the Technology Acceptance Model (TAM) to elaborate the determinants of the adoption of SEO practices. By TAM, the adoption of technology is mainly influenced by two elements, namely, perceived usefulness and perceived ease of use (Alalwan, 2020). Within the SMEs context, managers tend to pursue SEO strategies more when

they feel that the adoption of the techniques can greatly enhance business performance and when they feel that the use of the tools and techniques of SEO are relatively simple to use.

The conceptual framework, therefore, assumes that the adoption of SEO depends on the perceptions of usefulness and the ease of the use by the managers and that in turn, has an impact on digital marketing ability, SME competitiveness, and the overall firm performance. This framework will offer the systematic means of examining the contribution of SEO to the competitiveness of SMEs in digital markets, by combining digital marketing strategy with technology adoption theory.

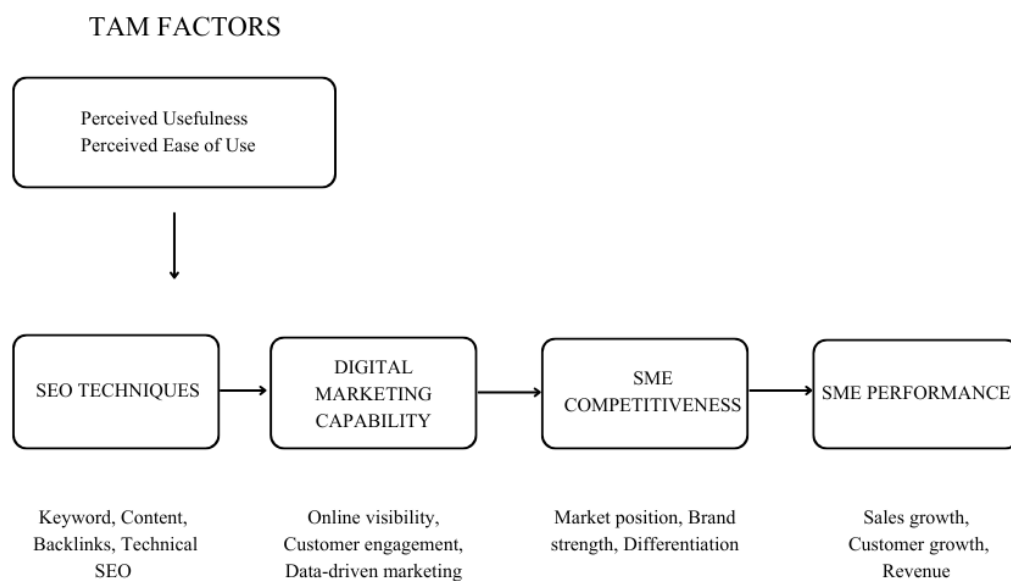


Figure 3: *Conceptual framework of SEO adoption and SME performance.*

Source: Developed by the author based on Davis (1989), Ahmad et al. (2024), Sun et al. (2022), Scuotto et al. (2021), and Kumar et al. (2023).

Figure 3 gives the conceptual framework of the study. This model has a hypothesis that the factors of Technology Acceptance Model influence the use of SEO technique which are perceived usefulness and perceived ease of use. Effective implementation of SEO will enhance competence of digital marketing and consequently the competence of the SME which ultimately leads to high performance of firms.

2.10 Digital Marketing Capability in SMEs

The digital marketing capability is the new strategic asset of small and medium-sized enterprises (SMEs) that must operate in the ever more competitive digital markets. With the transfer of the business to the online environment, the skill of efficient application of digital marketing tools, data analytics, and customer engagement strategies has become one of the primary factors of firm competitiveness and performance. Digital marketing capability is defined as the capacity of the firm to implement online platforms, data-driven marketing, and use of digital technologies to create value, facilitate customer relationship, and improve performance of the business (Kumar et al., 2023; Sun et al., 2022).

In the case of SMEs, the ability to do digital marketing is even more significant because such companies usually have fewer resources than large enterprises do. In contrast to big companies that have a considerable budget on marketing their products, SMEs need to use cost-effective digital tools to advertise their services and products. SMEs can connect with more people with the aid of digital marketing platforms like search engines, social media networks, and online content platforms without the need to invest a lot of money. With the creation of a robust digital marketing presence, SMEs will be able to become more visible in the market, be in a better position to establish better relationships with their customers, and compete better in online contexts (Scuotto et al., 2021; Quinton et al., 2021).

The capability to analyze and utilize information to make marketing decisions is one of the most significant components of digital marketing capabilities. Digital analytics can enable companies to gather and assess data pertaining to client interaction, web traffic, engagement habits, and campaign effectiveness. These understanding can give organizations the ability to make more focused marketing initiatives and can streamline their marketing campaigns dynamically. The ability to make marketing decisions based on data also enables SMEs to learn more about the preferences of customers and the market trends to create more efficient marketing strategies and enhance customer experiences (Sun et al., 2022). Companies that apply analytics properly can keep improving their marketing strategies and eventually achieve a better marketing performance and competitive advantage.

Content marketing is also essential to building the digital marketing capabilities. The idea of content marketing is that valuable, relevant, and consistent content should be developed and disseminated to help attract and maintain a well-defined audience. In the case of SMEs, the content

marketing can involve blog posts and educational articles, videos and social media material that contains useful information to the prospective customers. Good content enhances internet presence and contributes to search engine optimization (SEO) through relevancy and authority of websites. Kumar et al. (2023) explain that those companies that invest in content marketing tend to reinforce their brand position and gain organic traffic to their websites with the help of search engines.

Customer engagement is another important aspect of digital marketing capacity. The digital platforms offer SMEs first-line channels of communication with their customers, which can make them communicate with them in real time. Through the social media platforms, email marketing and other interactive features on the websites, firms would be able to interact with the customers, respond to queries, and establish long term relationships. Close customer engagement assists companies to develop brand loyalty and enhance customer satisfaction that eventually leads to better business performance (Scuotto et al., 2021). SMEs can generate a personalized marketing experience and enhance their competitive edge by ensuring constant communication with customers using digital channels.

Innovation and learning within organizations are also facilitated by the digital marketing capabilities. With the help of digital platforms and data analytics, companies will be able to gather valuable data regarding customer needs and the development of the new market trends. This knowledge can help SMEs to invent new products, services and marketing strategies that will match the expectation of the customers more. Competitors, who constantly use digital data and customer feedback to enhance their operations, stand a higher chance of changing in accordance with the shifting market realities and retaining their competitive advantage in the long-term (Sun et al., 2022).

Moreover, when digital marketing capabilities can be integrated with other organizational resources, they can contribute to an improved performance of firms to a considerable degree. The resource-based view of the firm explains that competitive advantage occurs when companies create special resources that are valuable, difficult to replicate, and in strategic relationship with business goals (Khin and Ho, 2019). Such digital marketing competencies as SEO experience, analytics, and content production abilities can be strategic assets that can make SMEs outperform their rivals in digital marketplaces.

Nonetheless, the establishment of digital marketing potentials among SMEs is not a smooth sail. A few small companies do not have the technical skills and finances needed to adopt the

progressive digital marketing technologies. SMEs also find it a challenge to apply various digital tools and platforms to a coherent marketing strategy. Despite these issues, companies investing in digital marketing education, technology usage, and data-driven decision-making have a higher chance to build their marketing potential and enhance the efficiency of the business overall (Quinton et al., 2021).

In general, digital marketing capability is important in helping SMEs to compete favorably in the digital markets. Through the data analytics, content marketing strategies and customer engagement platforms, SMEs can improve their online presence, customer relationship, and sustainable competitive advantage. The digitalization of marketing capabilities is a factor that will hold importance in the success of SMEs in the long term as digital technologies keep changing to enable them to develop and integrate.

Table 1: *Digital Marketing Capabilities in SMEs*

Capability	Description	Impact
Content marketing	Creating blog posts, educational articles, and digital content to attract audiences	Improves SEO ranking and brand visibility
Data analytics	Using digital analytics tools to monitor website traffic, customer behavior, and marketing performance	Enables data-driven marketing optimization
Social media integration	Using social media platforms to communicate and interact with customers	Enhances customer engagement and brand awareness
SEO capability	Optimizing websites and digital content for search engines	Increases organic traffic and online visibility
Customer engagement tools	Email marketing, chat systems, and interactive platforms	Strengthens customer relationships and loyalty

2.11 SEO and Customer Acquisition

Search engine optimization (SEO) is nowadays one of the most significant digital marketing techniques that companies involved in gaining customers within online settings adopt. With the continued dependence of consumers on search engines like Google to get products, services and information, businesses need having their websites ranked highly in the search engine results page (SERP) lists. SEO allows the companies to enhance the visibility of their web pages by optimizing web page information, structure, and technical aspects to ensure that the information that is important to a search engine can be readily identified and ranked by a search engine. Better positions in search results have a significant likelihood of attracting potential customers to the site of a firm and use the services it provides (Berman and Katona, 2019).

SEO is of great benefit to small and medium-sized enterprises (SMEs), as in this case, it offers a low-cost method of acquiring customers as opposed to the conventional methods of advertising. In contrast to the paid advertising campaigns where the company must engage in financing the campaign, SEO means creating organic traffic by enhancing the position of the company in the search engines. Organic traffic is the number of visitors to a site who come up unpaid in the search results as opposed to paid adverts. Such traffic is regarded as one of the most valuable as it demonstrates the real interest of the users who are currently searching products or services (Ahmad et al., 2024).

Among the key advantages of SEO, there is the fact that it helps businesses to create leads. Lead generation is a process that entails the identification of potential customers that are interested in the products or services offered by a company and might end up becoming paying customers. Upon searching the search engines using relevant keywords, and finding a business site in the search results, users can open the site, browse through its pages, and be involved with what the firm has to offer. Such interactions are possible leads that companies can cultivate using the digital marketing tactics like email marketing, targeted promotions, or individual communication (Kannan and Li, 2021).

SEO sensitivity in generating leads is best associated with the relevance and quality of the contents of websites. Advanced algorithms of search engines are used to rank the relevance of web pages with references to the use of keywords, quality of content, user interaction, and the authority of the site. Companies that produce good content that responds to customer questions and needs have higher chances of ranking high in the search engine listings. Consequently, content marketing and

SEO are highly interrelated, where the usefulness of material does not only attract the user but also enhances the visibility of a site in search engines (Järvinen and Taiminen, 2020).

SEO generated organic traffic has long-term advantages to SMEs. SEO can provide a long-term traffic to the site, as opposed to paid advertising campaigns where the campaign ceases when the advertising budget runs out. When a site has been ranked high on the appropriate keywords, it can still maintain its visitors even without the need to incur extra marketing costs. Such prolonged presence is what makes SEO a long-term investment with SMEs aiming at achieving long-term operations in digital markets (Berman and Katona, 2019).

The other significant benefit of SEO is that it increases customer trust and credibility. Consumers tend to see sites which come up at the first position on the search engine results as more trustworthy and authoritative. The search engines focus on sites that supply good, reliable, and useful content and customers are likely to rank higher sites better. Consequently, the improved SEO results of the business can lead to the improvement of brand-trust and consumer-confidence (Saura et al., 2021). In digital markets, trust is a vital aspect to consider since the online customers cannot see and touch the products or services they are purchasing in the physical world. Hence, companies cannot afford to use digital signals like search engine ranking, web design, customer commentaries, and content quality to create the confidence of prospective clients. SEO also helps in this process of building trust as it enhances the overall user experience of websites. In the contemporary world of SEO, the focus is on the speed of loading, responsiveness to mobile devices, easy access to navigation, and focused content because these elements make users more satisfied and willing to spend more time on the site (Sun et al., 2022).

Search engine optimization is also important in customer acquisition relying on the choice of keywords. Using key word research helps businesses to determine the search terms that are used by their potential customers when they seek a particular product or service. By focusing on these keywords in web materials, the businesses will be able to make sure that their web pages will feature in the relevant search results. To be more precise, long-tail keywords, which are more specific and less competitive search terms are also highly effective when it comes to SMEs. These are the keywords that can bring smaller amounts of traffic but usually, lead to increased conversion rates as they represent more specific customer intentions (Ahmad et al., 2024).

Besides providing a customer attraction service, SEO may also be used in customer retention and relationship building. The more the users keep seeing a brand in search engine results in their

search of relevant information, the more familiar they will be with the brand, and the more inclined they will become to rely on it. This exposure is repeated which helps in brand recognition and relationship with customers in the long run. These relationships can be reinforced by digital marketing strategies that integrate both SEO and content marketing and social media engagement, and are able to deliver constant value to the customers (Scuotto et al., 2021).

Moreover, SEO would help to strengthen the competitiveness of SMEs in online markets. When compared to smaller organizations, larger organizations usually have a significant advertising budget that enables them to conquer the conventional marketing media. Nonetheless, SEO gives SMEs a chance to compete in a better manner by addressing niche markets, particular keywords, and quality content. SMEs can earn good positions in the search engines and capture highly targeted audiences despite their financial limitations by being the best in certain fields and producing valuable content (Kraus et al., 2021).

Nevertheless, effective SEO strategies are difficult to implement and need to be undertaken on a constant basis and strategy. Search engine algorithms are constantly being modified to give better search results and eliminate any form of manipulation. Companies should therefore constantly check their performance in search engines, refresh their content, and change their strategies so that they remain high in search engines. Companies, which use a more proactive approach to SEO and align it with the larger digital marketing initiatives, stand a higher chance to become successful in digital marketplaces in the long run (Kumar et al., 2023).

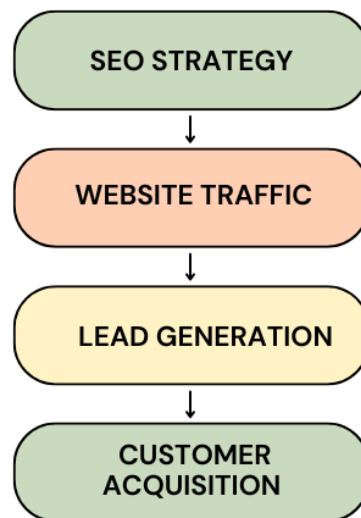


Figure 4: SEO Strategy and Customer Acquisition Process

Source: Berman & Katona, 2019; Järvinen & Taiminen, 2020; Saura et al., 2021

2.12 Conceptual Model and Hypotheses Development

Considering the literature surveyed in the prior sections, this research paper identifies a conceptual model, as follows: based on the theory, there is an interrelationship between the adoption of search engine optimization (SEO), digital marketing capability and competitiveness in SMEs, and performance of the firm. The model incorporates the digital marketing research, the resource-based view (RBV), and the Technology Acceptance Model (TAM) to determine how digital marketing practices lead to the strategic success of small and medium-sized enterprises (SMEs). All these theoretical approaches imply that the competitive advantage and the performance results of the companies in digital markets may be improved because of the introduction of digital marketing technologies and the empowerment of organizational capabilities (Khin and Ho, 2019; Kumar et al., 2023).

The modern markets have changed the way the firms communicate with their customers, advertise their products, and provide value through digital technologies. It is crucial that SMEs use digital marketing tools, as they offer a relatively affordable channel of reaching their customers and competing with large institutions. One of these tools has become a vital part of the digital marketing strategies, as SEO has a direct impact on the online presence, the number of visits to the site, and customer interaction. Companies that manage to integrate SEO principles can increase their online presence and become more effective in attracting potential clients to the organization by means of organic search ranking (Berman and Katona, 2019; Ahmad et al., 2024).

The effect of SEO to the performance of the business is not in a vacuum though. In its place, SEO is helping to build the expanded digital marketing capacities in companies. Digital marketing capability is the skill of organizations to apply digital technologies, data analytics, and online tools in an efficient way to meet marketing goals and generate customer value. Companies that build their digital marketing would be more able to study customer behavior, to create specific marketing campaigns, and to streamline their marketing activities (Sun et al., 2022).

The conceptual framework of this paper thus supposes that the adoption of SEO leads to the establishment of digital marketing capability, which in turn affects the competitiveness of the SMEs and the performance of the firms. This paradigm also recognizes the fact that SEO can have a direct effect on the business performance in terms of raising the traffic levels of the webpage, enhancing brand awareness, and attracting new clients.

SEO Adoption and Digital Marketing Capability

The relationship under the first investigation in the current study is associated with the impact of the adoption of SEO on the digital marketing capability. SEO techniques including keyword optimization, content development and web optimization demand the firms to have analytical capabilities, digitalized tools, and marketing expertise. With the adoption of the strategy of search engine optimization, companies become more proficient in handling online space, examining the work of websites, and learning about the history of search activities of customers. The activities also lead to the advancement of larger digital marketing capabilities that improve the capacity of the firm to compete in digital markets.

Past studies indicate that the greater the adoption of digital marketing technology by firms, the more they tend to build data-driven marketing capabilities and enhance their marketing performance (Kumar et al., 2023; Scuotto et al., 2021). SEO adoption can thus be used as an access point to enable SMEs to enhance their digital marketing capabilities. According to this argument, the hypothesis below is suggested:

H1: SEO adoption positively influences digital marketing capability in SMEs.

Digital Marketing Capability and SME Competitiveness

The competence of digital marketing is important in promoting competitiveness of SMEs. Companies that are well equipped in digital marketing will be well positioned to know the trends in the market, communicate with the customers using digital platforms and provide customer-centric marketing experiences. The capabilities enable the firms to stand out among others and leverage their market power.

Digital marketing technologies can also help companies to obtain useful information on customer preferences and market dynamics. This knowledge enables SMEs to change their marketing approaches and come up with new products and services that address customer needs to a greater extent. The resource-based view suggests that such organizational capabilities as digital marketing expertise may be used as a strategic resource that helps firms to attain sustainable competitive advantage (Khin and Ho, 2019).

As empirical studies have shown, the digital marketing capabilities have a strong contribution to the firm competitive power and innovation performance (Scuotto et al., 2021; Quinton et al., 2021). Companies, which successfully apply digital marketing technologies, would be better placed to

capture the customers, increase their markets, and build their brands in the digital markets. Consequently, the hypothesis is as follows:

H2: Digital marketing capability positively influences SME competitiveness.

SME Competitiveness and Firm Performance

Competitiveness is the capacity of a firm to be in a strong place in the market compared to the other firms in the market. Competitiveness to SMEs can be easily determined by how the firm is able to differentiate its products or services, how the firm is able to save its customer relationships and how the firm is able to react to the changes in the market. The greater the level of competitiveness, the more likely the firm can have better financial performance, acquisition of customers as well as growth in business.

Digital marketing skills can make the company competitive, as they will help the company to increase brand awareness, better customer communication, and overall market growth. With the SMEs getting a firmer grip on the digital competitiveness, they can receive better performance-related results, including more sales, retention, and increased brand awareness (Sun et al., 2022; Kumar et al., 2023).

Past research on the topic of digital marketing and strategic management has repeatedly established a positive correlation between the competitive advantage and the performance of the firm (Wamba et al., 2020). Companies with highly competitive advantage are at a better position to record high performance than their rivals. To this end, the hypothesis below is put forward:

H3: SME competitiveness positively influences firm performance.

SEO Adoption and Firm Performance

Besides the indirect effect of SEO implementation on the performance of the firm via the digital marketing capability and competitiveness, it can also directly affect performance. SEO enhances the visibility of the websites in the search engine ranking and this allows the businesses to get potential customers who are actively searching in the search engine regarding the product or services that the business is offering. More traffic to the site will result in more leads, better customer response and eventually more sales and revenue.

In the case of SMEs that have smaller marketing budgets, SEO is an especially useful tool of enhancing the effectiveness of marketing. In comparison to the conventional advertisement procedures which demand huge financial input, the process of SEO allows companies to create organic traffic and long-term presence without having to spend on advertising on a regular basis

(Berman and Katona, 2019). Because of this, those SMEs investing in SEO might have a better marketing result and a more robust business performance.

Empirical evidence indicates that companies that have profitable SEO strategies tend to claim greater online exposure, greater web traffic, and a higher rate of getting customers (Ahmad et al., 2024). Thus, the usage of SEO can have a direct impact on the performance of business as it can reinforce digital appearance of the firm and engage new clients. According to this argument, the hypothesis below is suggested:

H4: SEO adoption positively influences firm performance.

Table 2: Hypotheses of the Study

Hypothesis	Relationship
H1	SEO adoption → Digital marketing capability
H2	Digital marketing capability → SME competitiveness
H3	SME competitiveness → Firm performance
H4	SEO adoption → Firm performance

3. Methodology

3.1 Research Philosophy

Research philosophy is the core of assumptions that regulate the research process and may include assumptions about the nature of reality (ontology), the nature of knowledge (epistemology), and the methods that should be used to research social phenomena (Saunders et al., 2019). Research philosophy forms the basis of defining research methods, data collection procedure, and data analysis method. The research philosophy is a particularly significant consideration in the field of research studies about digital technologies and business performance since it defines the interpretation of relations between technology adoption and the performance of organizations.

The positivist research paradigm is popular in studies that are based on digital marketing technologies and firm performance. Positivism presupposes that the reality is not dependent on the researcher and objective knowledge could be acquired in terms of empirical observation and statistical analysis. Research under this paradigm seeks to establish quantifiable correlations

among variables and to establish theoretical hypotheses using quantitative data. It is the positivist approach that would best fit the study of causal relationships between the adoption of technologies and the final performance of firms (Kraus et al., 2021; Maroufkhani et al., 2022).

The research philosophy used in this study was positivism, as the main aim was to explore the nature of relationship between search engine optimization (SEO) adoption, digital marketing capability, SME competitiveness, and performance of firms. Such relations are measurable objectively with references to numerical data gathered in the organizations and evaluated with the help of statistical data. The positivist paradigm enables the researcher to develop hypothetical assumptions based on the previously existing theoretical frameworks and empirically test them by data.

The other motivation behind the adoption of a positivist philosophy in the research is the application of available theoretical models like the technology acceptance model (TAM) and digital capability theory. Quantitative methods are usually applied to test these theoretical frameworks and enable researchers to test the relationships between technological adoption and organizational outcomes. In particular, the measurable constructs considered by TAM include perceived usefulness and perceived ease of use, which are most analyzed with the help of structured survey tools and statistics (Alalwan, 2020; Chatterjee and Kar, 2020).

Positivism research methods have also been used in previous empirical researches exploring the digital transformation and adoption of digital marketing in SMEs. As an illustration, to research the connection between digital capabilities and firm performance, the theoretical models are often tested using quantitative survey data to find statistically significant relationships between variables (Scuotto et al., 2021; Kumar et al., 2023). These researches show that positivist research models can be successfully used to explore how the use of technology can affect organizational competitiveness and performance.

Table 3: Research Design Summary

Element	Choice
Research philosophy	Positivism
Research approach	Deductive

Research strategy	Survey
Methodological choice	Quantitative
Time horizon	Cross-sectional
Data type	Primary data
Data collection method	Structured questionnaire
Sampling technique	Purposive sampling
Target population	SMEs
Data analysis techniques	Correlation and regression

3.2 Research Approach

This research was deductive research. The deductive method starts with the already existing theories and conceptual frameworks and subsequently comes up with hypotheses that are empirically established based on the amassed data (Saunders et al., 2019). Deductive reasoning is a logical process whereby facts are explained using theoretical ideas.

The deductive research method was suitable in this study since the research was based on theoretical views that were previously established regarding the concept of digital marketing, digital transformation, and technology adoption. The theoretical model created during the literature review suggested that the application of SEO increases the digital marketing capacity, which, in turn, raises the competitiveness of SMEs and the performance of firms. Such correlations were based on past research studies which analyzed the issue of digital capability formation and organizational performance (Khin and Ho, 2019; Vial, 2019).

In the deductive approach, the research initially used a literature review to determine important constructs and theoretical connections. Based on these theoretical understandings, a conceptual framework was designed to understand the role of the adoption of SEO in the competitiveness and performance of SME. Empirical data based on a structured survey of SMEs was then used to test the relationships that were suggested in the framework.

The application of deductive research is prevalent with empirical research studies on the scope of the effects of digital technologies on the performance of firms by enabling the researcher to test the relationship between theories through statistical techniques. Using such statistical methods as correlation analysis and regression analysis, the researchers may conclude whether the suggested relationships between variables are empirically validated (Trainor et al., 2020; Sun et al., 2022).

The other benefit of the deductive method is that the researcher can test his or her hypothesis based on theories. In the research, theoretical constructs of TAM and the theory of digital capabilities were applied to explain the role of managerial perceptions and organizational capabilities on the adoption of SEO and the performance of digital marketing. The application of these theoretical relationships to empirical data will help to give an insight into the applicability of these theories in the context of SME digital marketing.

3.3 Research Strategy

This study used a form of quantitative survey research to respond to the research objectives. Survey research has been extensively employed in research that has looked at the adoption of digital marketing and its effect on organizational performance since it enables a researcher to gather standardized information on a large population of respondents. The use of surveys allows measuring perceptions, attitudes, and behaviors in a variety of organizations, and thus it is appropriate when studying patterns and relationships among variables (Saunders et al., 2019).

The survey plan enabled the researcher to acquire data on the SEO usage of SMEs, their digital marketing abilities, and the perceived business performance. Through the data gathered in different companies, the research could determine of recognizing general tendencies and exploring associations between digital marketing activities and the performance of a company.

Research based on surveys is especially widespread in the research that explores the issue of digital transformation and adoption of marketing among SMEs. Since SMEs are in the various industries and various organizational settings, surveys offer an effective way of gathering similar information to be used in various firms. Past studies exploring the relationship between digital adoption and business performance have often used survey data to conduct research on the relationship between SME digital marketing capabilities and performance (Odoom et al., 2020; Quinton et al., 2021).

The other strength of survey research is that it is suitable in measurement of constructs which were developed based on theoretical frameworks like TAM. Perceived usefulness and perceived ease of use constructs are generally quantified with the help of structured questionnaires, which include the attitude of the respondent to the technology adoption (Alalwan, 2020). Survey procedures thus offer an appropriate means to operationalize the theoretical constructs and find out their relationship with organizational performance.

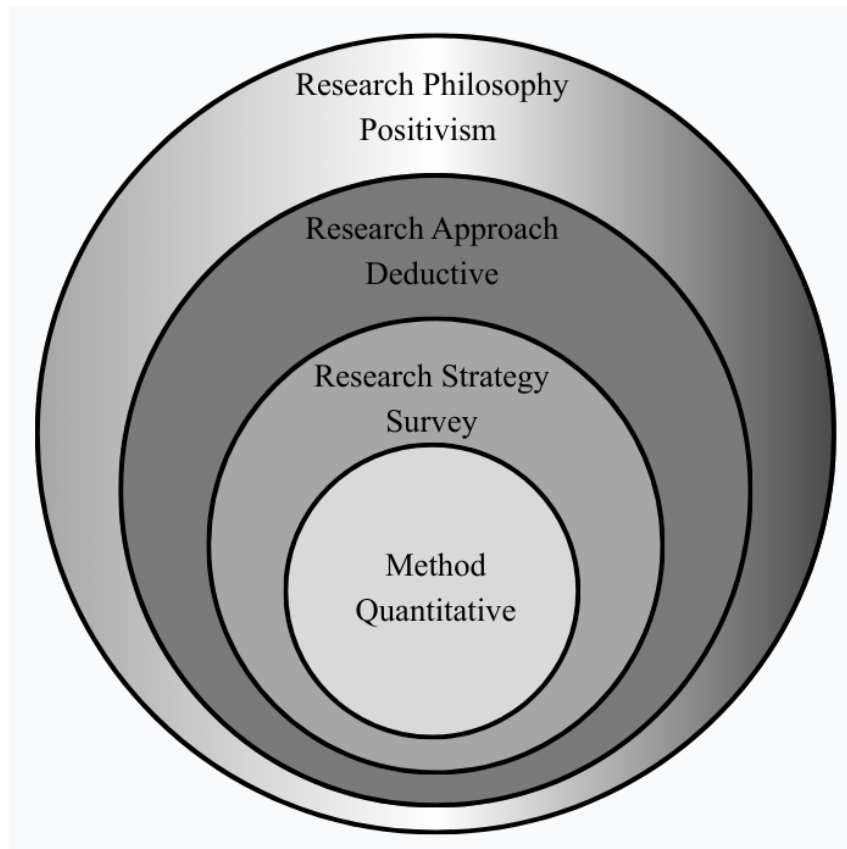


Figure 5: Research onion illustrating the methodological design of the study.

Source: Adapted from Saunders et al. (2019).

The results of figure 5 confirm that the methodological design of this research is presented by the framework of the research onion. The study adopts the positivist research, deductive research approach, survey method, and quantitative research as research philosophy in the data collection and analysis.

3.4 Sampling and Target Group (SMEs)

This study targeted to include small and medium-sized enterprises (SMEs) that are quite active in using digital marketing tools in carrying out business activities. The unit of analysis was the SMEs since they have special challenges and opportunities in embracing digital marketing technologies and competing in the digital markets. SMEs have fewer resources than large corporations usually and, therefore, must focus on the cost-efficient marketing approach to stay competitive (Kraus et al., 2021; Müller et al., 2021).

A purposive sampling was utilized to select the respondents who have the valid knowledge on the digital marketing activities of their firms. Purposive sampling is a non-probability sampling process that enables the researcher to sample those subjects with certain information or experience pertaining to the subject of the research. The respondents were chosen in the framework of this study due to their participation in the process of digital marketing and SEO decision-making in their organizations.

The SME owners, founders, marketing managers, and digital marketing specialists were the respondents. These were deemed as the right people since they are the ones who design and implement digital marketing strategies. Their opinions were very informative about the implementation of the SEO practices, the obstacles that were encountered in the process, and the perceived effects of the SEO on the business performance.

The reason why purposive sampling is commonly used when conducting research on SMEs is that the sample should be comprised of respondents who have the knowledge required to give meaningful answers to questions that were posed in the questionnaire. Purposive sampling has also been applied in previous researchers to investigate digital transformation and marketing adoption among SMEs to include respondents with topical managerial or technological experience (Gavrila & de Lucas Ancillo, 2021; Eggers et al., 2020).

One hundred and fifty valid answers were gathered, and they were analyzed statistically. This was deemed adequate sample size to carry out quantitative analysis, which will involve correlation and regression analysis. Sample sizes of 100 to 200 respondents are usually used to conduct organizational research to analyze variables interrelationship and test theoretical models (Maroufkhani et al., 2022).

Table 4: *Target population and sample*

Criteria	Description
Target population	Small and medium-sized enterprises (SMEs)
SME definition	Based on the European Union classification (European Commission, 2020)
Respondents	Owners, founders, and marketing/digital marketing managers
Industry scope	Multiple sectors (e.g., services, retail, manufacturing, e-commerce)
Geographical scope	[Country / region of study]
Sampling method	Purposive sampling
Sampling technique	Non-probability sampling
Sample size	Valid responses
Unit of analysis	Individual SME

3.5 Data Collection

The primary data used in this study were obtained by means of a structured online questionnaire which has been given to the owners of SMEs and marketing managers. Online surveys are common in online marketing research since they enable a researcher to obtain data that concerns geographically spread respondents in an efficient and cost-effective manner (Trainor et al., 2020).

The questionnaire was also administered online via the Internet resources and professional networks, where the respondents needed to be actively engaged in digital marketing practices in their organizations. The purpose of the research was explained to the respondents and their responses were guaranteed to be confidential and anonymous.

The completed responses were 150 valid responses that were used as final dataset in statistical analysis. Analysis was done by first sifting out responses that were either incomplete or not consistent. The received data helped to gain an idea about the scale of adoption of SEO by SMEs,

methods of implementation of SEO, and perception of the effects of SEO on competitiveness and business performance.

The theoretical framework and measurement scales were developed using secondary data acquired in academic literature in addition to primary data. Past empirical research on the adoption of digital marketing and the performance of SMEs was used to design the questionnaire and determine the constructs to be relevant in measuring (Kumar et al., 2023; Sun et al., 2022).

3.6 Questionnaire Design

The questionnaire in the current study is formulated based on the validated measurement scales of the previous studies of digital marketing capability, the level of technology adoption, and firm performance (Kumar et al., 2023; Sun et al., 2022). The reliability and validity of the research instrument is maximized using standard scales of measurement.

The questionnaire had five major sections. The former section involved the gathering of firm demographic data, such as the industry in which the firm operates, its size, and time of existence. These variables were identified to explain the nature of the sample and analyze how the digital marketing adoption can vary among different organizational settings.

The second part was the measurement of digital marketing adoption which concentrated on the utilization of digital tools and platforms to market the product. The third section discussed the practices of SEO, such as optimizing the keywords, the development of the content, technical SEO, and the strategy of the backlinks.

The fourth part involved scale measures that were based on the Technology Acceptance Model such as perceived usefulness and perceived ease of use of SEO technologies. These were added constructs to examine the impacts possessed by managerial perceptions to the adoption of SEO practices among the SMEs.

The last measurement was the SME competitiveness and firm performance, with the measures of customer acquisition, online presence, and business growth.

The measures of all responses were taken based on a five-point Likert scale, beginning with strongly disagree (1) to strongly agree (5). Likert scales are often applied to online marketing

research practice since they enable researchers to measure the level of perception and attitudes of respondents in relation to the use of technology (Alalwan, 2020).

Table 5: Questionnaire structure and measurement sources

Section	Variables / Constructs	Measurement scale	Source
Firm demographics	Firm size, industry, years of operation	Categorical	Self-developed
SEO adoption	Extent of SEO use, frequency of optimization activities	5-point Likert scale	Ahmad et al. (2024)
SEO techniques	Keyword, content, technical SEO, link building	5-point Likert scale	Ahmad et al. (2024)
Perceived usefulness (PU)	Impact of SEO on traffic, leads, and sales	5-point Likert scale	Alalwan (2020)
Perceived ease of use (PEOU)	Ease of learning and implementing SEO	5-point Likert scale	Alalwan (2020)
Digital marketing capability	Online visibility, customer engagement, analytics usage	5-point Likert scale	Sun et al. (2022)
SME competitiveness	Market position, brand strength, differentiation	5-point Likert scale	Kraus et al. (2021)
Firm performance	Sales growth, customer acquisition, business expansion	5-point Likert scale	Kumar et al. (2023)

3.7 Data Analysis Methods

This study used quantitative statistical analysis to obtain results of the survey of small and medium-sized enterprises (SMEs). Quantitative analysis enables the researcher to test the association between variables based on numerical data and statistic methods. The method is common in the

literature that examines the adoption of digital marketing and its effect on firm performance since it assists researchers to evaluate the hypothesized relations and make objective conclusions on the premise of empirical data (Saunders et al., 2019). Statistical analysis of the study was in the form of Statistical Package used (SPSS) software that is widely applicable in social science and business research as a data management and statistical testing.

The descriptive statistics were used in the analysis of the first phase to summarize the characteristics of the sample and give a general overview of SEO adoption among SMEs. Descriptive statistics assists researchers in knowing the underlying characteristics of the data by providing the data in form of frequencies, percentages, means and standard deviations. In the given research, the descriptive statistics helped to characterize the demographic features of the involved companies such as industry type, firm size, firm age, and respondent position. This move gave broad information about the sample and was used to distinguish the trends in the use of SEO among SMEs. The descriptive analysis plays a significant role in survey-based studies since a clear picture of the data set are obtained before more sophisticated statistical methods are used.

The reliability analysis was the second phase of the analysis to determine the internal consistency of the measurement scales applied in the questionnaire. Reliability can be defined as the degree to which a measure should give consistent and stable measures in relation to varying measurements. Reliability in this research was tested by use of Cronbach alpha coefficient that is one of the most common measurements of internal consistency in quantitative research. Significant values of Cronbach alpha (above 0.70) are typically regarded as to be acceptable, which means that the measurement items employed to identify a specific construct are both reliable and consistent (Sun et al., 2022). The key constructs to be utilized in this study, such as SEO adoption, digital marketing capability, SME competitiveness, and firm performance, were subjected to the reliability test. One of the key measures was to ensure that the measurement scales were reliable, to undertake additional statistical analytical processes.

After the reliability test, correlation analysis was performed to test the relationships of the key variables that were used in the conceptual framework. Correlation analysis indicates the strength and the direction of a relationship between two variables. In the study, correlation coefficients were performed using Pearson correlation coefficients to establish the relationship between the adoption of SEO, digital marketing competency, SME competitiveness, and firm performance. The

correlation analysis assisted in establishing the presence of positive relationships between these variables and the ability to support the findings of the theoretical assumptions offered in the conceptual model. This is because correlation analysis is often applied in the digital marketing and technology adoption studies to understand how organizational capabilities and performance outcomes relate to each other (Maroufkhani et al., 2022).

The last phase of statistical analysis was the regression analysis; this was done to investigate the effects of SEO adoption and the digital marketing capability as it relates to the competitiveness and firm performance of SMEs. Regression analysis is a statistical method of research which enables the researcher to analyze the influence of one or more independent variables on a dependent variable. This article employed regression models to establish whether adoption of SEO and digital marketing capability make significant contribution to the achievement of competitiveness and performance by SMEs. The analysis was able to provide information about the strength and significance of such relationships by estimating regression coefficients.

Regression analysis can be strongly applied in empirical research addressing the issue of the effect of digital technologies on the organizational performance as it allows researchers to test theoretical hypotheses and measure the effect of various factors (Wamba et al., 2020; Maroufkhani et al., 2022). Reliability analysis, correlation analysis, and regression analysis could help this study to analyze the relationships between SEO adoption, digital marketing capability, SME competitiveness, and firm performance comprehensively. These statistical methods gave strong empirical data to justify the research objectives and conceptual model of the research.

3.8 Reliability and Validity

Quantitative research requires the reliability and validity of the measurement tools applied since reliable and valid research findings would be obtained. Reliability is the stability and constancy of a measuring tool, that is, the measuring tool should give similar results when measuring the same phenomenon under the same condition. Validity on the other hand is the degree to which the measurement instrument measures accurately the concept that it is supposed to measure. That is, validity is a guarantee that the research tool can capture the actual meaning of the constructs under investigation. Reliability and validity are also valuable in survey-based studies since they can be used to ensure that the data obtained has been properly reflective of the theoretical constructs (Saunders et al., 2019).

The reliability of the study was measured in terms of internal consistency of the measurement scales that were involved in the questionnaire. Internal consistency is the extent to which various items, which assess identical construct, yield similar outcomes. To measure internal consistency Cronbach alpha coefficient was employed. One of the most popular reliability measures of the social science and digital marketing research is Cronbach's alpha. It measures how close the items in a scale are to each other and whether they are gauging the same concept. The Cronbach alpha of 0.70 or more is an acceptable value according to widely accepted research standards and represents that the scale of measurement is reliable (Sun et al., 2022).

The reliability analysis of this study was performed on the major constructs incorporated in the conceptual framework of this study, which are SEO adoption, digital marketing capability, SME competitiveness, and firm performance. The outcomes of the reliability test made it clear that all the constructs obtained the Cronbach's alpha value exceeding the recommended level of 0.70. These findings affirm that the measures of the constructs were internally consistent and can serve to continue statistical analysis. It was also necessary to attempt ensuring internal consistency since the study incorporated several items in the survey to measure such complicated phenomena as digital marketing capability and organizational competitiveness.

Besides reliability the validity of the research instrument was also put into great consideration. Validity makes sure that the measurement scales are correctly reflecting theoretical constructs applied in the research model. Constructing validity of quantitative studies is one of the key elements of the validity since it is defined as the degree to which the measurement items are related to the theoretical constructs that they are designed to measure. Construct validity is especially significant when the study deals with abstract variables like technology adoption, digital marketing capability, and firm performance.

To achieve construct validity, the measurement scales employed in this research were borrowed and modified on past validated undertakings on digital marketing, technology adoption, and performance of SME. The fact that scales to be used are the same scales that have been previously tested in other empirical studies, aids in ensuring that the survey tool measures what it is supposed to measure. As an example, the measurement items connected with digital marketing capability and firm performance were borrowed and modified based on the existing research in digital marketing (Kumar et al., 2023; Sun et al., 2022). The scales of measurement were previously

empirically validated in these studies and this is what enhances the validity of the constructs in the study.

Also, the fact that there was a fit between the scales of measurement and the conceptual framework also contributed to the validity of the research design. All the constructs in the questionnaire were directly connected to the theoretical relationships that were put forward in the conceptual model. This consistency made the data obtained by use of the questionnaire to be useful in analyzing the relationship between the adoption of SEO, digital marketing competence, SME competitiveness, and performance of the firm.

In general, the reliability and validity procedures used in the study contributed to ensuring that the measurement tool was consistent and accurate in the measurement of the most significant constructs of interest. The study employed Cronbach alpha rendering the internal consistency to evaluate it and validated measurement scales used in previous studies to ensure a high level of methodological rigor. These measures enhanced the validity of the data and offered good support on the consequent statistical measurements involved in testing the research hypotheses.

3.9 Ethical Considerations

Academic research has to include the use of ethical considerations especially when there are human subjects to be dealt with in the study. Ethical research practices are important because they guarantee the participants the right, privacy and well-being are preserved during the research process. A few ethical principles were also adhered to to carry out the research in a responsible way and ensure that the research process is not compromised in this study. Such principles were informed consent, voluntary participation, anonymity, confidentiality, and the responsible use of data (Saunders et al., 2019).

To begin with, there was voluntary participation in the process of data collection. The survey was fully voluntary and no one was pressured or coerced to take part in the study. The purpose and objectives of the study were explained to the participants before they could fill out the questionnaire to decide on whether to participate or not. This procedure assisted in maintaining that the respondents had a perfect understanding of what was the nature of the research and that they were participating voluntarily. Voluntary participation by the study participants is an essential

ethical consideration in academic studies because it honors the autonomy and the right to decide by people (Saunders et al., 2019).

Second, the participants gave their informed consent before filling in the questionnaire. The purpose statement was clear in the introductory part of the survey, as well as the nature of the information that was to be gathered and how the data would be utilized. The respondents were made aware that the study aimed at determining how small and medium-sized enterprises (SMEs) embrace search engine optimization (SEO) and digital marketing behavior to improve their competitiveness. This information was given beforehand and therefore the respondents were able to make a choice on whether they wanted to participate in the study or not.

The other ethical principle observed in this study was confidentiality. All the answers obtained out of the participants were considered confidential and were used only in academic purposes. No external organizations and individuals were given the information that respondents provided. The confidentiality of the participants should be considered especially when it comes to SMEs since companies might be worried about disclosing strategic data devoted to their marketing policies, digital policies, or business performance. Confidentiality will make the respondents give truthful responses without fearing that their information can be used against them.

Along with the issue of confidentiality, anonymity was also observed during the research process. The survey questionnaire did not ask about the personal identifiers as the names of the participants and the names of their companies. This did not allow the identification of individual respondents based on the data obtained. Anonymity also minimizes the chances of a bias in the responses as there are more chances that the respondents will give truthful answers since they realize that their identity cannot be linked to them. Anonymity will also ensure that there is no exposure of the participants to potential risks that can come because of disclosing sensitive business information.

Moreover, the information gathered during this research was placed in a safe location and could be utilized exclusively to conduct academic analysis as it applies to the research objectives. No commercial or non-academic use was done with the data. When the analysis was done, the information was managed in a responsible manner such that the privacy of the participants was not compromised. During the process of academic research, ethical management of research information is one of the most vital conditions since it helps to avoid the possibility of misusing or unauthorized access to sensitive data (Saunders et al., 2019).

In SME research, ethical issues are particularly significant as businesses usually provide strategic or operational data that in turn may cause a change in their competitive stance in the market. Thus, it is important to retain the trust of the participants to receive validated information and secure the validity of the research results.

All in all, the research followed the ethical principles of conducting social science studies. The research preserved the rights and privacy of research participants by guaranteeing voluntary participation, informed consent, confidentiality, anonymity, and sound data management that bypassed the ethical considerations of the research process. The reliability and credibility of the study were facilitated by these ethical practices and contributed to making sure that the research was conducted in a responsible and professional way.

3.10 Pilot Study

A pilot study was conducted to test the clarity, reliability, and the effectiveness of the research questionnaire in general before the actual survey was conducted. Pilot study is a pre-test research study that is done prior to the actual data collection process to test the research instrument, and find out the possible issues in the questionnaire design. A pilot study is important to researchers to make sure that the questions in the surveys are comprehensible, pertinent to the research goals, and can yield dependable data (Saunders et al., 2019). Pilot testing is also significant in the process of quantitative research since the researcher is given a chance to refine the structure and wording of the questionnaire to be given to a larger sample.

In this study, the pilot test was done on 10 SME managers and business owners who were engaged in digital marketing activities in their respective organizations. The selection of these participants was since they had working experience with the use of digital marketing tools, such as search engine optimization (SEO), social media marketing, and online interaction with the customer. They had experience to give useful feedback on the relevance and clarity of the survey questions. The pilot sample was selected among SMEs that work in the services sector, retail industry, and e-commerce as these are the areas of digital marketing practices.

The pilot questionnaire had the same format as the final survey instrument, which will be used in the actual research. It contained several sections that dealt with firm demographics, adoption practices of SEO, ability to perform digital marketing, perceived usefulness, and easiness of use of the SEO technologies, SME competitiveness, and firm performance. The questionnaire was

given to the respondents who were requested to fill it in and give their comments on how clear the questions were, how long the survey was taking, and how the constructs being measured were relevant.

The questionnaire was refined with the help of the feedback received after testing it on the pilot participants. A few respondents also reported that some of the technical terms used in relation to SEO and digital marketing were not explained. To overcome this problem, small adjustments were carried out to make the language of some questions easier to understand and to make sure that the terms employed in the survey was easy to comprehend by the SME managers with different degrees of digital marketing knowledge. The significance of this step lies in the fact that SMEs vary in terms of digital maturity and level of technological knowledge (Kraus et al., 2021).

The pilot study also assisted in testing the time that will be taken to complete the survey besides enhancing the clarity of the questionnaire. The pilot respondents stated that the survey took about 8-10 minutes to fill in and this was a reasonable time to take during an online survey. To maximize the extent of responding, it was necessary to maintain the survey not too long to allow the respondent to become less fatigued when the actual data collection process commenced (Trainor et al., 2020).

The pilot study also assisted in determining the reliability of the measurement scales to be applied in the questionnaire. Most questions in the survey were derived out of already validated measurement scales that had been applied in the research on digital marketing capability, technology adoption, and SME performance. The utilization of the established scales assists in increasing the reliability and the validity of the research instrument since the measures were previously tested in other empirical studies (Sun et al., 2022; Kumar et al., 2023). The answers of the pilot responses revealed that most of the questions in the questionnaire were consistent and were appropriate to gauge the constructs incorporated in the conceptual framework.

The other critical findings of the pilot study were the affirmation that the questionnaire form was logically structured and easy to navigate by the respondents. The questionnaire was modeled in such a way that the respondents responded to the questions regarding the characteristics of their firm and then proceeded to the questions regarding the digital marketing practices and perceptions regarding the adoption of SEO. The structure is what enabled the respondents to take their time and answer more detailed questions pertaining to the marketing practices and the performance of firms.

Generally, the pilot study was important in determining the quality of the research instrument before the complete survey was given out. The pilot testing of the questionnaire on a small sample of SME managers helped the researcher to determine the minor problems associated with wording, structure, and terminology and adjust it accordingly. These were to contribute to the reliability and readability of the final questionnaire that would be used in the process of data collection. The pilot study thus enhanced the methodological rigor of the study and the chances of developing credible and relevant data to be used in further statistical analysis of the results.

3.11 Data Screening

Data screening was done on the data collected before the main statistical analysis was performed. Data screening is a crucial part of quantitative research since it can aid in the process of verifying that the data is correct, has no missing data, and can be analyzed statistically. The steps include detection of the possible problematic data in the dataset, including missing values, outliers, and statistical assumption violations like normality. Avoiding such concerns prior to statistical analysis is beneficial to enhance reliability and validity of the research results (Saunders et al., 2019).

The initial process of data screening was the screening of missing data values. Missing data can be a result of a respondent not responding to all the questions in the survey or responding to some of the questions but leaving some of them unanswered. With many missing values, it is possible that they would impact the correctness of the statistical figures and decrease the credibility of the research. In this paper, the data was analyzed in SPSS to detect bad responses and absence of answers in the questionnaire forms. Significant incomplete responses were not to be analyzed to guarantee the quality of the data set. Questionnaires that were not filled in were not kept to undergo the final analysis; this left 150 valid responses to be used in the statistical analysis. Deleting insufficient responses aided in making the dataset complete and good enough to do additional statistical activities.

The second step consisted in determining the outliers in the data set. Outliers are those extreme values which do not match with others in the data set. These atypical values can be due to a mistake in data entry or measurement mistakes or to inconsistent responses given by the respondents. Outliers can misrepresent the results of the statistics because they have the potential to falsely affect the statistical estimates like the means, correlations, and regression coefficients. Thus, outliers' detection and assessment are a key element of data preparation in quantitative studies.

Descriptive statistics and graphical tools that are provided in SPSS including boxplot, standardized scores etc., were used to examine potential outliers in this research. Several observations that had an abnormally high or low value were cautiously checked to see whether these were real responses or data errors. As it was observed in the analysis, there were no extreme outliers that can radically change them. Hence, the data set contained all the valid responses to be analyzed further. The need to ensure the absence of problematic outliers is crucial in enhancing the strength and the validity of statistical procedures when conducting research studies that involve the use of organizational and marketing data (Maroufkhani et al., 2022).

The third data screening process entailed testing of the normality of data distribution. Normality is a concept used to depict how the data distribution is distributed in a normal (bell-shaped) shape. Various statistical methods applied in quantitative studies such as correlation analysis and regression analysis assume that the variables to be collected are normally distributed. In case the data largely deviates away to normality, this can impact the statistical results.

To measure normality, various measures were analyzed such as skew, kurtosis, and graphic analysis like histogram. Skewness is used to measure the distribution of the data in terms of symmetry, and kurtosis measures the extent of the peaked distribution as compared to a normal distribution. The skew and kurtosis values lie within acceptable limits, implying that the data distribution is not very different. The outcomes of the normality test showed that the variables applied in this research were almost normally distributed and therefore could be parametrically analyzed using statistical tests like Pearson correlation and regression analysis. Normality testing is highly advisable in any research involving the establishment of the relationship between digital technologies and firm performance since the test guarantees the satisfaction of statistical assumptions (Wamba et al., 2020).

In general, data screening process indicated the data was appropriate in subsequent statistical analysis. The analysis of the missing values used ensured that only the full responses were used in the analysis, the analysis of the outliers ensured that there were no extreme values that could potentially distort the results, and the normality test showed that the data were in a form to be used in the analysis as the parametric statistical methods presuppose. These initial data checks made the dataset reliable and valid and guaranteed that the further statistical analyses would include the

valid and valuable information concerning the relationship between SEO adoption, digital marketing capability, SME competitiveness, and firm performance.

3.12 Statistical Model

In this study, a statistical model consisting of regression was used to test the relationships as stated in the conceptual framework. The regression analysis is a widely used quantitative method, which enables the researcher to investigate the correlation between the independent variables and a dependent variable and eliminate the effect of other variables. Regression analysis is commonly applied in the scientific literature addressing the impact of digital capabilities and technology adoption on the performance of firms in terms of digital marketing (Maroufkhani et al., 2022; Wamba et al., 2020).

The regression model was used to test whether the adoption of SEO, digital marketing capability, and competitiveness of SMEs has any effect on the performance of firms in this study. These variables were obtained based on the conceptual framework that came up during the literature review. The model presumes that the introduction of SEO assists with the evolution of the digital marketing competency that, in its turn, makes SMEs more competitive and the basis of the ultimate firm performance.

A few benefits accompany regression analysis when it is to be used to analyze these relationships. Firstly, it allows the researcher to approximate the size and the direction of the effect of each of the independent variables on the dependent variable. Second, it enables one to test theoretical hypotheses put forward in the research model. Third, regression analysis assists in estimating whether the noted relationships among variables should be considered as statistically significant. These attributes render regression analysis to be quite appropriate when using empirical research to examine the effectiveness of digital marketing tactics on the competitiveness and performance of firms (Kumar et al., 2023).

The regression model in the current research aims at testing the impact of the key constructs on the SME performance outcomes. SME performance is the dependent variable in the model and it is the business results realized by the firm, in terms of customer acquisition, market expansion, and business success. The independent and mediating variables will be SEO adoption, digital

marketing capability, and the competitiveness of SMEs as the strategic capabilities that affect the firm performance.

The regression equation used in this study can be expressed as follows:

$$\text{SME Performance} = \beta_0 + \beta_1 (\text{SEO Adoption}) + \beta_2 (\text{Digital Marketing Capability}) + \beta_3 (\text{SME Competitiveness}) + \varepsilon$$

Where:

- β_0 represents the intercept of the regression model
- β_1 , β_2 , and β_3 represent the regression coefficients associated with each independent variable
- ε represents the error term capturing unexplained variation in firm performance

The regression coefficients also show how strong and in which direction each independent variable is related to SME performance. A positive coefficient implies that a rise in the independent variable is related to a rise in the dependent variable. As an example, when the coefficient of SEO adoption is positive and statistically significant, it means that the companies that embrace SEO activities have a greater chance of realizing better performance results.

This regression model was statistically analyzed using SPSS statistical software which is more common in business and social science studies to conduct regression analysis and other tests of statistics. The findings of the regression analysis are used to find out whether the hypotheses of the conceptual framework are attested by the empirical data.

Generally, the regression model offers a systematic methodology of exploring the effects of digital marketing strategies on the competitiveness and performance of SMEs. The model could provide useful information on how SMEs could use digital marketing strategies to enhance their competitiveness in the digital market by examining the role played by adoption of SEO and digital marketing capability in the performance of firms.

Table 6: *Variables Used in Regression Analysis*

Variable	Type
SEO adoption	Independent variable
Digital marketing capability	Mediating variable
SME competitiveness	Mediating variable
Firm performance	Dependent variable

Table 7: *Description of Variables in the Regression Model*

Variable	Description
SEO adoption	The extent to which SMEs implement search engine optimization practices to improve online visibility
Digital marketing capability	The firm's ability to utilize digital tools, analytics, and online marketing platforms effectively
SME competitiveness	The firm's ability to maintain a strong market position relative to competitors
Firm performance	Business outcomes such as customer acquisition, sales growth, and market expansion

4. Empirical Results

This chapter highlights the quantitative survey results of the research on the small and medium-sized businesses (SMEs). The key task of the given chapter is to study how SMEs have adopted search engine optimization (SEO), the types of the applied SEO strategies by the firms, the biggest obstacles connected with the adoption of the SEO, and its influence on the competitiveness and performance of SMEs.

The discussion is based on the conceptual framework introduced in the previous chapter and discusses the research questions pointed out in the introduction. To investigate the relationship

between the major constructs as suggested by the study (SEO adoption, digital marketing capability, SME competitiveness, and firm performance), descriptive statistics and correlation analysis were used to conduct statistical analysis.

4.1 Sample Characteristics

The sample of provided study contained 150 small and medium enterprises that operated in different industries. The sample of companies representing various industries will contribute to the fact that the results will be more representative of the SMEs involved in the digital economy. The respondents were comprised of business owners, marketing managers as well as digital marketing specialists directly engaged in making decisions regarding digital marketing and SEO strategies.

The inclusion of the respondents, who were managers and strategists, was a good measure to have the data that would have reflected the organizational views on the issue of adoption of digital marketing and business performance.

The dispersion of industries in the firms shows how SMEs are different participants in the digital markets. The findings indicate that 41.3 percent of companies are in the service industry and it is the industry that is most represented in the sample. Service-based business frequently requires a lot of digital marketing and online presence in the process of getting customers; hence the popularity of the businesses featured in the study.

The retail firms constitute 25.3 percent of the sample and 18.0 percent of the firms are in e-commerce. E-commerce presence is also especially pertinent since online-based businesses heavily rely on search engine presence and internet marketing tactics to bring clients on board. The manufacturing companies constitute 15.4 percent of sample, which indicates growth in the adoption of digital marketing practices in conventional industries.

The size of the organizations represents that most of the organizations surveyed are micro and small firms, which is predictable regarding population of the European SME sector. 42.7 percent of the firms are micro enterprises that have less than ten workers. Micro enterprises would tend to use low-cost digital marketing solutions like SEO since they do not have financial capacity to carry out a massive advertising campaign.

Small businesses that hire up to forty-nine employees constitute 35.3 percent of the sample, whereas the medium-sized firms that hire between fifty and two hundred and forty-nine employees

constitute 22.0 percent of the sample. The existence of companies in various classes of sizes enables the research to compare the differences in SEO usage and online marketing capacity in various organizational setups.

Concerning the firm age, the findings show that the sampled SMEs are a quite balanced representation of the levels of the business maturity. About 30.7 percent of the companies have existed under five years, which implies the availability of relatively new businesses that perhaps are still in the initial phases of adopting digital marketing. The 36.7% of the sample is comprised of firms that have been in operation between five and ten years, with the 32.6% of firms being in operation over ten years.

Table 8: *Sample characteristics (n = 150)*

Variable	Category	Frequency	Percentage (%)
Industry	Services	62	41.3
	Retail	38	25.3
	E-commerce	27	18.0
	Manufacturing	23	15.4
Firm size	Micro (<10 employees)	64	42.7
	Small (10–49 employees)	53	35.3
	Medium (50–249 employees)	33	22.0
Firm age	< 5 years	46	30.7
	5–10 years	55	36.7
	> 10 years	49	32.6
Respondent role	Owner / Founder	82	54.7
	Marketing manager	44	29.3
	Digital marketing specialist	24	16.0

The table 8 presents the characteristics of the SMEs that were included in the study. Most of the companies belong to the service sector (41.3), retail sector (25.3), e-commerce (18.0), and manufacturing (15.4). The micro enterprises are the highest percentage (42.7) in terms of firm size which is not surprising considering the make-up of European SME sector. Out of the sample, 35.3 is composed of small companies, and the medium-sized enterprises comprise 22.0. Regarding the age of the firms, most of the SMEs lie between five and ten years (36.7), which is quite mature business population with an established market experience. Owners and founders (54.7%), then marketing managers (29.3%), then digital marketing specialists (16.0), is the general type of respondent, which implies that the data were collected with the respondents having first-hand knowledge about the digital marketing and SEO operations.

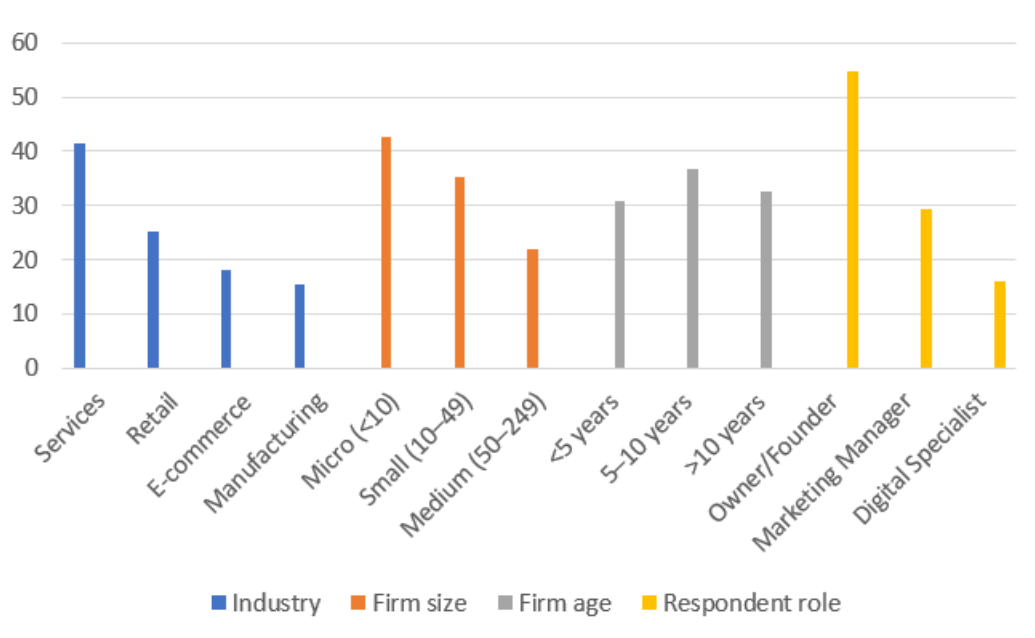


Figure 6: *Distribution of sample characteristics.*

Most of the sample is service-sector SMEs as Figure 6 shows and this is characteristic of the design of the European digital economy where service-based businesses are more dominant in the online space. Micro firms constitute the largest population that demonstrates the importance of low-cost digital marketing platforms such as SEO to small businesses with low resources. The distribution of firms age is comparatively equal, which is a pointer that not only young SMEs are engaged in digital marketing, but also the established ones. In addition, the fact that the high

percentage of owner-managers among the respondents is a user level is also critical to ensure that the data received are strategic organizational viewpoints and not operational viewpoints.

4.2 Level of SEO Adoption

The empirical evidence reveals that a significant part of SMEs is implementing SEO as a digital marketing campaign. Nonetheless, the degree of adoption of SEO also differs greatly amongst the firms, which can be explained by the disparity in digital capacity, management experience, and access to resources.

There are cases of SMEs using SEO as the middle part of their digital marketing strategies and incorporating it into more comprehensive marketing endeavors like content marketing and customer engagement campaigns. These companies usually use a strategic approach to SEO, constantly optimizing the content in their websites, checking the positions in search engines, and assessing the data on the traffic of their websites.

On the contrary, other SMEs do not have a complex SEO strategy, but use less advanced SEO techniques, including the use of simple keyword optimization or simple web indexing. Such companies are usually unable to cover the resources or knowledge needed to conduct more sophisticated optimization methods.

The difference in the level of SEO-adoption is attributable to the Technology Acceptance Model (TAM). As it is described in TAM, the adoption of a specific technology depends on the perceived ease of use and perceived usefulness (Alalwan, 2020; Chatterjee and Kar, 2020). SMEs that believe that SEO is a good tool in enhancing online presence and getting customers are more likely to invest in the implementation of SEO.

It is also found in the results that the firms with stronger digital marketing possibilities prefer more advanced practices of SEO. The SMEs that have a greater degree of digital capability are more likely to adopt data analytics tools, do keyword research, and be aware of the measures of web performance. The results are in line with the earlier studies indicating that digital capabilities are important factors in the improvement of organizational competitiveness and performance (Khin and Ho, 2019; Sun et al., 2022).

The findings also support the previously mentioned literature that suggests that SEO is one of the most frequently utilized digital marketing tools by the SMEs because it is relatively inexpensive and has long-term marketing advantages (Ahmad et al., 2024; Berman and Katona, 2019).

The findings displayed in Figure 6 show the level of adoption of SEO among the sampled SMEs. The moderate adoption group covers most firms, who have been using the basic search engine optimization methods, but not yet considered applying SEO to its strategic marketing planning.

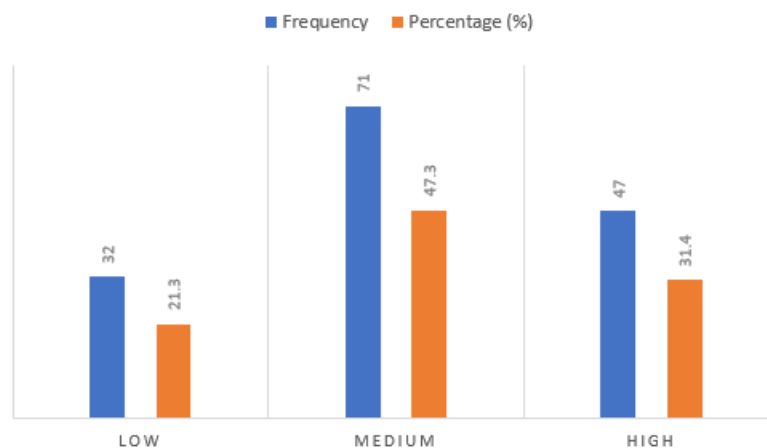


Figure 7: Level of SEO adoption among SMEs ($n = 150$).

Source: Survey data.

Figure 7 shows that most SMEs are in the moderate adoption level, which implies that they have already adopted the fundamental practices of SEO but have not fully incorporated SEO in their totality marketing practices. Many of the firms report high levels of adoption with low levels of SEO adoption by SMEs representing various levels of digital marketing maturity.

4.3 SEO Techniques Used

SEO analysis shows that the main aspect of SEO adopted by SMEs is on-page practices that are mostly of optimization of keywords and development of content. The reason why these techniques are used extensively lies in the fact that they are comparatively available and do not consume as many monies as complex SEO techniques.

One of the most common techniques that are undertaken by SMEs is keyword optimization. Companies usually determine the keywords that are related to their goods or services and use them in the websites, description of their products and even in their blogs. By using the appropriate

search words, the SMEs can have a higher probability of being listed in the search engine results when the potential customers are using the search engines to find corresponding products or services.

The use of content marketing is also central to the use of SME SEO strategies. Numerous companies constantly create blog releases, articles, and information content aimed at appeals to online crowds and better search engine ranking. Content marketing assists companies to gain authority in their respective markets and offer useful information that would boost customer interaction.

These results align with the prior studies that emphasize the role of quality of content in the promotion of a search engine position and brand positioning (Järvinen and Taiminen, 2020; Ahmad et al., 2024).

On the contrary, SMEs implement off-page SEO strategies, including the development of backlinks, less often. The construction of quality backlinks can be a strategic alliance, and it is not easily accessible to small companies due to specific expertise.

Equally, technical SEO, such as optimization of the speed of the websites, mobile responsiveness and use of structured data are more commonly employed by SMEs which are more digitally mature.

The average value of 3.47 shows that the adoption of SEO among the SMEs can be described as moderate which implies that the number of firms that find it necessary to adopt this type of optimization is large, yet there is a weak adherence to better optimization methods.

Table 9: *Descriptive statistics for SEO adoption*

Variable	Mean	Standard Deviation (SD)
Keyword optimisation	3.72	0.88
Content optimisation	3.85	0.81
Technical SEO	3.21	0.93
Backlink development	3.08	0.97

Overall, SEO adoption	3.47	0.76
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The findings in Table 9 show that the most popular of the SEO practices used by SMEs are content optimization (M = 3.85) and keyword optimization (M = 3.72). Technical SEO (M = 3.21) and backlink development (M = 3.08), in turn, are applied less often. Overall, the average value of 3.47 points at the moderate degree of SEO usage among the SMEs surveyed.

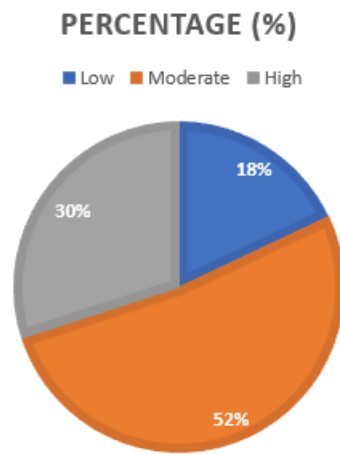


Figure 8: *Distribution of Digital Marketing Capability among SMEs*

Source: Survey data analysis (n = 150) Kumar et al., 2023; Sun et al., 2022; Scuotto et al., 2021

The distribution of digital marketing capability by SMEs is presented in figure 8. The bulk of the firms are moderate in terms of digital capability (52%), meaning that they use digital marketing tools simply. High capability (30 percent) is demonstrated by most SMEs who have stronger commitments to analytics and digital strategies. Conversely, 18 percent of companies are low in digital capability, which implies low utilization of digital marketing technologies. These findings demonstrate that there are differences in the level of digital maturity in SMEs.

4.4 Challenges in Implementation

Although the advantages of SEO are acknowledged, the SMEs are faced with several difficulties in applying the SEO strategies.

Resource limitations are the greatest problem that is found in the analysis. The lack of financial resources prevents hiring special SEO professionals or investing in sophisticated digital marketing instruments in many SMEs. The limitation generally compels companies to stick to simple SEO measures instead of using extensive optimization measures.

The findings can be compared with prior studies that mentioned resource constraints as one of the greatest barriers to digital transformation among SMEs (Kraus et al., 2021; Eggers et al., 2020).

The other significant barrier as reported by respondents is the technical knowledge and digital skills. The technical side of the process, such as search engine algorithms, searching keywords, and web analysis, are not known to many SME managers and employees.

Such knowledge gap hinders the efforts of SMEs to review the effectiveness of SEO strategies, as well as quantify the effectiveness of the invested digital marketing activities.

Also, the constant fluctuation of search engine algorithm poses a challenge to SMEs aiming at being top ranked in the search engine. The ranking algorithms of search engines keep on being updated to enhance the quality of search results and avoid manipulations. The SMEs are also incapable of adapting fast to these changes due to the lack of resources and expertise.

Table 10: Challenges in SEO implementation ranked by mean score ($n = 150$)

Rank	Challenge	Mean	Standard Deviation
1	Lack of SEO expertise and skills	4.12	0.79
2	Limited financial resources	3.98	0.83
3	Lack of time to implement SEO	3.87	0.86
4	Frequent changes in search engine algorithms	3.74	0.91
5	Difficulty in measuring SEO effectiveness	3.69	0.88
6	Lack of management support	3.21	0.95

4.5 Impact on Competitiveness

The findings of the empirical research indicate that there is a moderate positive correlation between the use of SEO and the competitiveness of SME.

Companies that integrate the use of SEO measures state increased amounts of traffic to the web sites, better customer attraction, and enhanced positioning in the market places. SEO helps SMEs to increase their online presence and high ranking in search engines to access their customers in a more efficient way and in competition with bigger companies in digital marketplaces.

Such results rely on the prior studies suggesting that the digital marketing potential is important to enhance the performance and competitive edge of firms (Kumar et al., 2023; Wamba et al., 2020).

SEO also helps build brand credibility and trust of the customers. Websites positioned at the top of search engine results are perceived by consumers as very reliable and authoritative at times. This means that the search placement can affect the buying behavior and enhance customer relationships (Saura et al., 2021).

In addition, the analysis indicates that those SMEs that develop SEO into a more comprehensive digital marketing approach record better performance results than other companies where SEO is taken as an independent strategy. This observation indicates the need to synchronize SEO strategies with the general digital marketing programs like content marketing, social media marketing, and data analytics (Scuotto et al., 2021).

Table 11: Correlation matrix of key constructs ($n = 150$)

Variable	1	2	3	4
1. SEO adoption	1			
2. Digital marketing capability	.648**	1		
3. SME competitiveness	.592**	.711**	1	
4. SME performance	.567**	.689**	.734**	1

Note: $p < 0.01$

The correlation analysis indicates that there is a positive and significant correlation between SEO adoption and digital marketing capability ($r = .648$), SME competitiveness ($r = .592$) and firm performance ($r = .567$). There is also a positive relationship between digital marketing capability and competitiveness ($r = .711$) and firm performance ($r = .689$).

The correlation coefficient between the SME competitiveness and the firm performance ($r = .734$) is the strongest which means that firm performance outcomes have been associated with stronger competitive positions. These results are empirical evidence of the conceptual model that has been put forward in this research.

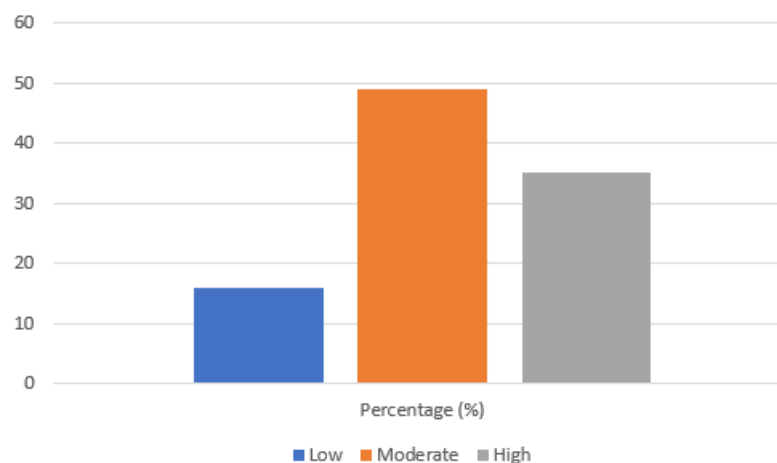


Figure 9: *SME Competitiveness Perception*

Source: Survey data analysis (n = 150) Kumar et al., 2023; Wamba et al., 2020; Scuotto et al., 2021

The perceived competitiveness of SMEs in digital markets is in Figure 9. The findings show that most firms view their competitiveness in the middle (49%), which means that they have a stable position in the market yet must contend with a high level of competition. High competitiveness is an aspect reported by around 35% of SMEs, which represent greater digital capabilities and marketing strategies. On the contrary, 16 percent of companies feel that they are not competitive enough, which means that they are struggling to embrace digital marketing tools. These discoveries demonstrate the role of digital capabilities and search engine optimization in enhancing the competitiveness of SMEs.

4.6 Reliability Test (Cronbach's Alpha)

Reliability analysis is a significant procedure applied in quantitative research to determine the consistency of measurement tools. Reliability is a term used in reference to the extent to which a group of measurement items should consistently gauge the same construct. SEO adoption, digital marketing capability, SME competitiveness, and firm performance are all examples of constructs that are measured on multiple items in survey-based research. As a result, there is a need to guarantee that these items yield similar results in case of measurement of the same concept. Cronbach alpha coefficient can be viewed as one of the most popular statistical measures to test internal consistency reliability (Sun et al., 2022).

The alpha of Cronbach is used to determine the level of correlation among individual items of a scale. A high value of Cronbach alpha illustrates that the items used in the scale measure the same concept as such they are highly consistent internally. Social science and management researchers usually accept a Cronbach's alpha of 0.70 or more to prove reliability, whereas values above 0.80 demonstrate high reliability (Kumar et al., 2023). Reliability analysis is necessary to have the measures of the constructs contained in the research model are accurate and constant.

Reliability analysis in this study was conducted by use of SPSS statistical software to assess internal consistency of constructs used in the questionnaire. It conducted an analysis based on four variables, namely, SEO adoption, digital marketing capabilities, SME competitiveness, and firm performance. All the constructs were measured on several items in the questionnaires, which are based on the previous scales that have been validated by the old literature. The reliability test was the one that studied the consistency in the answers to these items and the fact of their adequacy to further statistical analysis.

The outcome of the reliability test was that all constructs met Cronbach alpha value that was above the recommended value of 0.70. This proves that the measurement scales of the study were reliable and internally consistent. Particularly, the alpha-value of the Cronbach used in the adoption of SEO was 0.84, which means the reliability of the items that assess the degree of adoption of SEO practices by SMEs was high. The outcome of this indicates that survey questions that addressed the adoption of SEO effectively reflected the idea of the digital marketing optimization in the sampled firms.

In the same way, the value of the alpha of the digital marketing capability was 0.87 which is a very high degree of internal consistency. This outcome suggests that the items that assessed and dealt with digital marketing competence, such as the use of analytics, online presence, and involvement with customers were strongly interrelated and well-suited to assess the digital marketing competence of SMEs.

The reliability coefficient of SME competitiveness was 0.82 and this is also above acceptable level which signifies high reliability. This implies that, the questions in the questionnaires about the competitive positioning, brand strength and market differentiation were always indicative of the competitiveness of the firms involved.

Lastly, the alpha of the Cronbach used in the determination of firm performance was 0.85 which means that it has a high degree of reliability. This finding supports the fact that the items that signified the business outcomes like the customer acquisition, sales increase, and business expansion were internally homogeneous and could be used to measure the performance of SMEs.

The general findings of the reliability analysis reveal that the measurement scales applied in this research were valid and could be used in further statistical analysis. The values of Cronbach alpha are very high meaning that the items in the questionnaires were successful in measuring the constructs used in the conceptual framework. The development of good reliability should be especially relevant in the research of digital marketing strategies and organizational performance since reliable measurement tools can be used to secure the validity and trustworthiness of research results (Maroufkhani et al., 2022).

In general, the reliability test showed that the data obtained using the survey tool were coherent and applicable in further statistical analysis, including correlation and regression analysis. The reliability test enhanced the methodological quality of the study and provided the guarantee that the associations between the adoption of SEO, digital marketing ability, SME competitiveness, and the performance of the firm could be studied safely.

Table 12: Reliability Analysis (Cronbach's Alpha)

Variable	Cronbach's Alpha
SEO adoption	0.84
Digital marketing capability	0.87
SME competitiveness	0.82
Firm performance	0.85

4.7 Regression Analysis

To investigate the associations that have been suggested in the conceptual framework, the regression analysis was performed to identify how the adoption of SEO, digital marketing ability and competitiveness of SMEs affect the performance of the firms. The reason why regression analysis is popular in empirical research is the fact that it allows researchers to assess the impact of the independent variables on a dependent variable and what is more important is that regression analysis can be used to determine the strength and significance of the relationships. Regression models are also commonly employed in research on the effect of digital capabilities on firm competitiveness and business performance in the context of digital marketing (Maroufkhani et al., 2022; Wamba et al., 2020).

The dependent variable in this research was firm performance and SME competitiveness, Internet marketing ability, and SEO adoption were considered as independent variables. The SPSS statistical package was used to perform the regression analysis, and this enabled the estimation of the regression coefficients and the level of significance of each variable in the model. This analysis was aimed at testing whether the key constructs that are identified in the conceptual framework have a significant impact on the performance of SMEs.

The regression output is shown in Table X. The table will give the standardized beta coefficients (0), the strength and direction of the relationship between each of the independent variables and firm performance, and the corresponding p-values, which give the statistical significance of the relationships.

Table 13: Regression Analysis Results

Variable	Beta	p-value
SEO adoption	0.42	0.001
Digital marketing capability	0.37	0.003
SME competitiveness	0.48	0.000

The regression analysis outcomes indicate that the impact of all the three variables is positive and statistically significant on firm performance. The p-values of all variables are not more than the generally accepted value of significance of 0.05, meaning that the relationships that are found in the data set are significant and they are likely not contributed by chance.

The coefficient of regression concerning the adoption of the SEO ($= 0.42, = 0.001$) shows that there is a positive correlation between the adoption of SEO practices and performance of the firm. This finding indicates that SMEs that intensively apply the techniques of SEO will record improved business performance than those companies, which fail to effectively apply SEO. The positive coefficient means that the increase of online visibility and customer acquisition occurs with the help of the better practices in SEO, key-word optimization, recreation of the content and optimization of the web-site, and lead to the improvement of the performance of the firm. This result agrees with the existing studies on the relevance of SEO in enhancing digital marketing performance and business development (Berman and Katona, 2019; Ahmad et al., 2024).

The results of regression also indicate that digital marketing capability ($= 0.37, p = 0.003$) has a strong positive effect on the performance of firms. Digital marketing capability is the capacity of a firm in terms of its capacity to use digital technologies, analytics tools, and online marketing platforms in accomplishing marketing goals. The positive coefficient indicates that the better the digital marketing capabilities of the SMEs, the higher are the likelihoods of attaining greater levels of business performance. Companies with well-executed digital analytics, customer interaction systems, and data-based marketing are more likely to win and maintain customers in digital competitive markets. This result correlates with the previous literature that showed that the digital

marketing ability is a critical factor to enhance the performance of firms and their innovation performance (Kumar et al., 2023; Sun et al., 2022).

In the list of variables provided in the model, SME competitiveness ($= 0.48, p = 0.000$) shows the most significant impact on the performance of firms. The beta coefficient is comparatively higher, which shows that the firms with stronger competitive positioning have a better outcome in performance. Competitiveness stipulates how a firm can distinguish its services, the cordial relations that the firm is in with the customers and how a firm reacts well to market dynamics. Those SMEs that effectively develop competitive advantages due to digital marketing strategies and customer interaction can be expected to enjoy greater sales growth, better market share, and enhanced brand name. This observation aligns with the strategic management literature, which focuses on the significance of competitive capabilities in the organization performance drive (Khin and Ho, 2019; Wamba et al., 2020).

The regression analysis is thus in support of the theoretical assumptions in the conceptual framework. The findings indicate that SME competitiveness using SEO and the ability to market digitally leads to enhanced performance of the firm. These results underscore the strategic role of digital marketing practices in facilitating competence of SMEs in digital markets.

In addition, the findings underline that SEO is not supposed to be considered as a one-dimensional marketing tool but an instrument of strategic ability that helps to make digital marketing effectiveness more comprehensive. SMEs that incorporate the strategy of SEO in their digital marketing mixes and invest in the development of digital marketing skills have a higher probability of attaining long-term competitive advantages and better business performances.

All in all, the regression analysis gives a good empirical support on the relationship suggested in the conceptual model. The results show that these trends are very crucial and important because of the use of SEO, digital capabilities of marketing and competitiveness of SMEs determine their performance outcomes. Such findings confirm the fact that is emerging in the literature that digital marketing strategies are key determinants of the competitiveness and success of firms in digital economies (Maroufkhani et al., 2022; Kumar et al., 2023).

5. Discussion

This chapter is a discussion of the study empirical findings with respect to the research questions and the theoretical framework which has been constructed along the preceding chapters. The findings are discussed within the framework of available literature resources on digital marketing, search engine optimization (SEO), digital transformation, and the competitiveness of SME. The discussion also reviews the contribution of the empirical findings to the theoretical views presented in the literature review and its implications on the SME managers and policymakers.

The aim of this chapter is not to summarize the findings but also critically analyze the meaning and significance of them. Comparison of the results with past studies and theoretical models, the chapter gives more insight into the process of adoption of SEO as a digital marketing strategy by SMEs and the effects of the adoption on competitiveness and performance in digital markets.

5.1 Interpretation of Findings

The results of the present research point to the fact that the search engine optimization (SEO) is one of the key factors that help SMEs become more competitive in online markets. The active participants in the SEO practice are more likely to have a better online presence, higher traffic rates to their websites, and better customer acquisition opportunities. The above findings indicate that SEO is a valuable digital marketing instrument that companies can use to consolidate their market standing in a progressive digitalized world.

Among the major discoveries of the research is that there is a positive correlation between the adoption of SEO and firm competitiveness. Companies that adopted more SEO had higher rates of digital marketing opportunity and better business results. The result reinforces the overall thesis that the ability of digital marketing is an essential force in the performance of the firm in the digital economy (Kumar et al., 2023; Scuotto et al., 2021). Customers can be attracted by the ability to be visible in search engines and thus the power to win clients in online markets that are highly competitive can be very decisive of business success.

Online presence as a competitive tool is also important as noted in the results. The search engines have become the leading source of information to consumers who require products or services. Companies that feature in the first pages of search engines have chances of receiving customers

and leads. Thus, by using the application of SEO strategies, SMEs can easily compete with bigger companies enhancing their online presence and brand awareness.

The other key observation is that the adoption of SEO is widely different among the SMEs. Some of the firms have completely adopted SEO as part of their online marketing platforms but others are using simple methods of optimization. The difference herein is the diversity in terms of digital maturity, managerial skills, and organizational ability of SMEs.

It is observed that more advanced companies are more likely to engage in more advanced SEO practices. Such companies have a higher likelihood of using data analytics instruments, systematic key-word research and constantly measuring web site performance indicators. Consequently, less digital-enabled firms tend to use less sophisticated optimization techniques like placing keywords on the site or infrequent updates of the material.

This finding confirms that the process of digital transformation is multidimensional and incorporates changes on technological, organizational, and strategic levels (Verhoeff et al., 2021; Vial, 2019). The more successful SMEs which manage to introduce the digital technologies into their functioning are more likely to use the digital technologies as the means of marketing successfully.

It is also shown in the findings that the most popular SEO methods used by the SMEs are content marketing and keyword optimization. The practices are also quite simple to adopt and do not need much financial inputs thus they are especially appealing to small companies that have a limited marketing budget.

More sophisticated strategies, including technical SEO and building of backlinks, are, however, less widely used. The methods demand expertise in technical aspects and alliances that may not be easily accessible to smaller businesses. This observation indicates the structural resource constraints that the SMEs are usually exposed to in the process of embracing the high-level digital technologies.

In earlier research, the same authors indicated that another challenge faced by SMEs is the lack of technical skills and digital skills when adopting digital marketing strategies (Kraus et al., 2021; Müller et al., 2021).

The other notable observation of the research is that SEO is another factor that adds to competitiveness not only by generating more traffic of the websites but also by enhancing brand credibility and customer trust. High ranking in search engine is noted to be related with reliability and quality among the consumers. As a result, companies that are listed in the first search positions might enjoy an increased degree of customer confidence and interaction.

This finding indicates the strategic value of SEO in value creation to customers and brand positioning. SMEs will be able to build a better reputation and better relationships with customers by giving them pertinent content and enhancing their visibility on search engines.

The research also finds that there are several obstacles that prevent successful SEO implementation. The most described issues are the absence of digital skills, financial constraint, and the doubt on the efficiency of SEO strategies.

These issues highlight the significance of managerial consciousness and digital leadership in ensuring the success of the digital transformation in the SMEs. Companies that take active digital initiatives and invest in the formation of digital capabilities can overcome these obstacles and gain sustainable competitive advantages even more (Quinton et al., 2021).

5.2 Comparison with Previous Studies

The results of this research are mostly comparable to the earlier studies that investigate the connection between the digital marketing adoption and the SME performance. The previous research revealed that digital marketing technology results in the better performance of firms due to the increased customer interaction, market penetration, and facilitation of innovation (Sharabati et al., 2024; Maroufkhani et al., 2022).

The findings affirm that the SMEs which practice digital marketing can attract customers more and maintain their presence in the market. Nonetheless, the research contributes to the literature since it pays attention to the position of search engine optimization as a strategic instrument of digital marketing.

Whereas most past studies consider digital marketing adoption as an overall notion, the current study identifies the significance of personal digital marketing practices like SEO in promoting competitiveness and the success of the business. The study also offers a more precise insight into

how digital marketing affects the performance of firms by focusing on SEO as an independent variable of other digital marketing tools.

The results obtained by the authors confirm the results of Berman and Katona (2019), according to whom the visibility of search engines is the key to attracting customers and earning money. Their study shows that companies that are ranked in higher positions in search engines have high chances of getting people to visit their websites and translating them into consumers.

Likewise, Ahmad et al. (2024) show that SEO is also essential in enhancing online brand positioning, which also is in line with the findings of the current study. Their study shows that companies that use systematic approaches to SEO can enhance their brand image and engage more customers.

This study is also in line with the general literature on SME digital transformation. According to the previous studies, the implementation of digital technologies improves performance of firms but requires strong organizational capacity and environmental factors (Kraus et al., 2021; Eggers et al., 2020).

Nevertheless, the current research paper builds on the previous studies by showing how SEO is one of the most effective digital marketing strategies of SMEs. Compared to other digital marketing tools, which need significant financial outlay, SEO enables smaller companies to enhance their presence in online markets with very low resources.

The obstacles set during this research are also in line with the findings of the past researchers. The most common challenges to the adoption of digital marketing by SMEs have been reported as resource constraints, the absence of technical expertise, and uncertainty about the payback period (Cichosz et al., 2020; Trainor et al., 2020).

5.3 Theoretical Implications

This research paper adds to several theoretical understandings in the digital marketing field, technology adoption, and SME research.

To begin with, the study builds the literature on digital marketing and SME performance by theorizing the concept of SEO as a strategic ability and not as a technical marketing device. The research that has been done in the past was mostly based on social media marketing or overall

digital marketing adoption (Odoom et al., 2020; Trainor et al., 2020). Discussing SEO, this paper has given a deeper insight into the contribution of individual digital marketing practices in the competitive advantage.

Second, the research adds to the literature concerning the digital transformation and development of capabilities. The results prove that the implementation of SEO is closely linked with the building of the digital marketing potential in SMEs. Companies with greater digital capabilities will have higher chances of having better SEO strategies and make them part of the wider business strategies.

This result contributes to resource-based view (RBV) that implies that competitive advantage can be created based on valuable and hard to imitate organizational capabilities (Khin and Ho, 2019). The digital marketing skills and capabilities such as the expertise and skills in SEO and analytics are strategic resources that can help firms compete favorably in the digital markets.

Third, the study empirically supports the use of the Technology Acceptance Model (TAM) in investigating the use of digital marketing within the SMEs. The findings show that the perception of the managers on the usefulness and ease of use of SEO are important factors in adoption.

The success of SEO practices depends on SMEs, and managers think that these technologies would boost business results and feel that SEO tools can be rather simple to use (Alalwan, 2020; Chatterjee and Kar, 2020).

Lastly, the research combines TAM with the digital capability concept, which shows that the adoption of technology is not a sufficient condition to enhance the performance of the firm. Rather, competitive advantage arises when there is an involvement of technology adoption coupled with complementary organizational capabilities and strategic alignment.

5.4 Practical Implications for SMEs

The practical implications of the findings of this research to SME managers, digital marketing practitioners, and policymakers are described to be several implications.

To managers of the SME, the findings indicate that SEO is a long-term strategic investment and not a short-term promotional tool. Sustainable search engine visibility is a process that needs constant effort which involves developing content, optimization of the Web site, and monitoring performance.

The emphasis of SMEs should be in the creation of high-quality content and enhancement of the user experience on the websites since they are significant in the rankings of search engines. Organic traffic and authority in an industry are more likely to be achieved by firms that keep generating valuable content on a regular basis.

Another factor that has been highlighted in the study is the need to create digital skills and capabilities among SMEs. Numerous companies have difficulties with adopting the latest techniques of SEO because of the lack of technical skills. SMEs can solve these challenges by investing in employee training programs or by working with digital marketing experts.

This is especially important as digital capabilities were found to be the most important determinants of firm performance in the digital economy based on the previous research (Sun et al., 2022).

Strategically, SMEs ought to incorporate SEO to other digital marketing processes, such as content marketing, social media marketing, and data analytics. Companies that integrate various digital marketing tools are likely to have excellent marketing performance and customer engagement (Scuotto et al., 2021).

Moreover, digital marketing should be approached by means of data collection by SMEs. Using the information provided in the site traffic and the customer behavior the firms will be able to constantly optimize their marketing approaches and enhance their position in the search engines.

The implications of the findings to the policy makers and organizations that offer business support are also important. The SME digitalization can be supported with the help of training, developing digital infrastructure and financial support which can help SMEs to tackle resource constraints and enhance their competitiveness.

5.5 Implications for Digital Strategy

The results of the present study have valuable implications on how the digital strategies can be developed within small and medium-sized enterprises (SMEs). With the growing impact of digital technologies in competitive markets in the contemporary markets, SMEs need to pursue holistic digital approaches, which incorporate multiple online marketing platforms, such as search engine optimization (SEO), content marketing, and data analytics. The empirical findings of this study prove that the companies that have successfully implemented SEO in their overall digital

marketing framework are more likely to gain better competitive standing and enhance their business performance. Thus, SMEs ought to view SEO as a technical undertaking, yet, not as the sole element of the overall digital marketing strategy (Kumar et al., 2023; Scuotto et al., 2021).

Among the implications is associated with the formulation of SME marketing strategies. The common methods of marketing in the traditional way are usually based on the offline marketing promotional activities, which are not only expensive but also not easily measurable as compared to the digital marketing media. On the other hand, digital marketing allows SMEs to attract specific audiences, monitor customer behavior, and make promotional activity performance assessments in real time. The research results of the current study opine that SMEs need to engage in the use of integrated marketing strategies that involve the use of SEO in conjunction with other marketing digital platforms including social media marketing, content marketing, and online advertisement. These combined strategies allow companies to achieve the maximum exposure in the Internet and increase their market share in the online space (Kannan and Li, 2021; Sun et al., 2022).

Moreover, the findings also put a strong emphasis on digital maturity in the effectiveness of the SEO strategies. Digital maturity is a quality of organization that has both the technological infrastructure, digital ability, and strategic focus to organize and execute digital technologies successfully. The more digitally mature SMEs are, the more likely they will employ more advanced methods of SEO, data analytics tools, and combine digital marketing efforts across multiple platforms. Such companies are more likely to show an improved marketing performance due to their ability to analyze customer data and react faster in case of a market adjustment. This finding reinforces earlier studies indicating that the digital transformation is a slow process that has to be supported by building organizational capabilities and managerial competences (Verhoff et al., 2021; Vial, 2019).

The other important implication is made in relation to the strategic nature of SEO as a long-term investment. In contrast to paid advertising, which creates a short-term but immediate presence, SEO has long-term effects of enhancing exposure on organic searches, as well as causing long-term traffic of the site. Companies investing in quality content, technical optimization of the websites, and regular use of keywords will be able to develop formidable online presence as time goes by. This would be maintained visibility that will lead to better brand awareness, customer loyalty, and better market performance. The empirical findings of this paper affirm the fact that

systematic SEO practices of SMEs lead to increased degrees of competitiveness and business performance. The results align with the recent studies that asserted the importance of SEO in digital marketing campaigns over long-term (Berman and Katona, 2019; Ahmad et al., 2024).

Also, SMEs are advised to adopt the value of data-driven marketing approaches to improve digital performance. In the contemporary digital marketing platforms, there is a wide variety of analytical tools that enable the firms to track the traffic, customer behavior, and the performance of marketing campaigns in websites. With these tools, SMEs can determine the best SEO techniques, to optimize contents as well as to allocate resources to marketing in a more efficient way. Decision-making that is based on the data helps the firms to constantly get better with their marketing approaches and change in line with the changing market conditions (Sun et al., 2022).

Strategically, capabilities development and acquisition of knowledge should also be a priority of SMEs in the field of digital marketing. The research found that there are various issues that were associated with digital skills and lack of resources that fail to enable SMEs to practice high-level SEO fully. To address these adversities, the companies are advised to invest their resources in the training schemes, digital marketing education, and partnerships with other digital marketing consultants. Increasing internal digital capabilities will allow SMEs to utilize digital technologies more effectively and have sustainable competitive advantages on the online markets (Kraus et al., 2021; Müller et al., 2021).

Altogether, the results of this study highlight that effective digital strategies need a comprehensive approach that involves the incorporation of technology adoption, development of capabilities, and alignment of the strategy. SMEs that consider SEO as a element of their digital marketing strategy, invest in their digital capabilities, and make decisions based on data have higher chances of realizing greater competitiveness and long-term business success in the digital economy.

5.6 Implications for Policy Makers

The results of this research also have significant implications to policymakers and institutions in general, which support the development and the competitiveness of small and medium-sized enterprises (SMEs). With digital technologies becoming a leading trend in the business environment, governments are also critical in supporting SME digitalization and enabling businesses to embrace digital marketing practices including search engine optimization (SEO).

The findings of the study point to the fact that even though SMEs are becoming increasingly aware of the strategic significance of SEO and online marketing, most companies are under serious constraints due to resource limitations, the absence of technical expertise, and the deficiency of digital skills. As a result, special policy measures are required to help SMEs to overcome these issues and utilize digital opportunities to their fullest extent (Eggers et al., 2020; Kraus et al., 2021).

Financial and institutional support of SME digital transformation is among the most significant aspects, in which policymakers can invest their efforts. The financial capacity of many SMEs is low, and they might be unable to invest in advanced technologies of digital marketing or analytics and professional training. Government initiatives to issue grants, tax benefits or even subsidized digital infrastructure can assist SMEs to invest in digital technologies and enhance their presence on the web. These efforts may motivate companies towards strategic digital marketing and improve their competitiveness in the digital markets. As discussed in previous literature, policy assistance is crucial to the process of empowering SMEs to go digital with the new technologies and enhance their output in the more digitalized economies (Muller et al., 2021; Scuotto et al., 2021).

The other policy implication that is imperative is connected with the creation of online training and capacity-building programs of SMEs. The empirical findings of the present research indicate that numerous companies have difficulties with adopting advanced forms of SEO because of the deficiency of technical skills and digital marketing literacy. Digital skills and marketing training, the SEO strategy, content marketing, and data analytics would be of great help in helping SMEs to leverage on digital tools effectively. Governments and business support organizations should also work with universities, technology providers, and industry associations to establish special training courses that would focus on enhancing SME digital capabilities. Enhancing the digital capabilities of SME managers and workers will accelerate the speed of the digital transformation and enhance the competitiveness of firms in the online markets (Sun et al., 2022).

Moreover, policymakers have an opportunity to influence digital awareness and knowledge sharing amongst the SMEs. Numerous small businesses are hesitant in terms of optimizing the benefits of digital marketing and the possible investment payoff related to the search optimization techniques. SMEs may be assisted to gain access to the information and best practices in digital marketing through such public programs as digital innovation hubs, knowledge-sharing platforms, and mentorship programs. Such platforms can also be used to cooperate with digital technology

professionals by allowing firms to gain insights into successful cases of digital transformation (Maroufkhani et al., 2022).

Moreover, the governments are advised to come up with national digital strategies, which ought to be SME specific. These strategies may involve policies that are geared toward enhancing digital infrastructure, broadband, and promotion of the use of e-commerce platforms and digital marketing tools. Better digital infrastructure is also relevant especially to helping SMEs to access wider markets and become more integrated in digital economies (Vial, 2019).

Generally, the findings of this paper indicate that policy support should be effective in helping SMEs to surpass the obstacles that relate to adoption of digital marketing. The policymakers can assist SMEs to build the capacity necessary to execute the SEO strategies and compete favorably in the digital market through financial support of digital training programs, and empowerment of digital infrastructure. These policy programs not only help individual companies but also support the growth in general economy, innovation, and employment in the SME sector (Eggers et al., 2020).

6. Conclusion

In this chapter, the authors summarize the primary findings of the study and indicate the theoretical and practical value of the research. The chapter also addresses the limitations of the study as well as suggests future research directions. This study was conducted to research on the application of search engine optimization (SEO) as a digital marketing tool by small and medium-sized businesses (SMEs) to enhance competitiveness and performance in digital markets.

6.1 Summary of Key Findings

The results of this research affirm that SEO is now a significant strategic digital marketing instrument to SMEs that are being forced to operate in more competitive digital marketing settings. The search engine optimization has emerged as one of the primary drivers of business success as companies continue to use digital platforms as the primary means of value creation and communication with buyers (Verhoeff et al., 2021; Vial, 2019).

The empirical research showed that SMEs use SEO at varying degrees of sophistication. Although there are companies that apply comprehensive SEO techniques that encompass content marketing, technical optimization, and performance monitoring, there are companies that use simple

techniques like the optimization of keywords. Such differences may be in large part attributed to the alterations in digital capabilities, managerial experience, and resource presence. The greater the digital capabilities of the SMEs, the higher the chances that they will incorporate the element of SEO into their overall marketing strategies and record better performance results. This result aligns with the existing literature that has proved that digital capability is one of the primary determinants of firm performance and innovation (Kumar et al., 2023; Sun et al., 2022).

The findings also show that content marketing and keyword optimization are the most popular SEO among SMEs. The techniques are also comparatively cheap and simple to use, and therefore, they can be applied to Companies with limited finances. Nevertheless, when it comes to more sophisticated methods, like the development of backlinks and technical SEO, the practices are less commonly adopted because of the lack of technical expertise and Internet abilities. This is helpful in proving that, resource constraints continue to serve as a significant obstacle to digital transformation among SMEs (Kraus et al., 2021; Müller et al., 2021).

The other significant result of the research is that there is a positive correlation between adoption of SEO and competitiveness of SMEs. Companies that are aggressive in their SEO activities have had increased traffic on their websites, getting more customers, their brands enhanced, and positioning better in the market. The results confirm that digital marketing capabilities play an important role in sustainable competitive advantage in digital markets (Scuotto et al., 2021; Wamba et al., 2020).

The research also supports the argument that the perceived usefulness and perceived ease of use are critical determinants of adoption of SEO amongst SMEs. The fact confirms the relevance of the Technology Acceptance Model (TAM) when it comes to the adoption of digital marketing. They are also more likely to use SEO when they view it as an effective means of enhancing the performance of business and when they have the knowledge and skills required to use it to a successful effect (Alalwan, 2020; Chatterjee and Kar, 2020).

6.2 Contribution to Knowledge

The study has several significant contributions to the literature on the topic.

To begin with, the paper builds upon the research on digital marketing and the performance of SMEs by analyzing SEO as a particular strategic asset instead of viewing digital marketing as one

comprehensive unit. Past research tended to concentrate on social media marketing or simply the use of digital marketing (Odoom et al., 2020; Trainor et al., 2020). This study will have a closer insight by breaking down SEO and finding out the contribution of each individual digital marketing tool to competitiveness of SMEs.

Second, the study can be discussed as contributing to the body of research in terms of digital transformation and capability building as it illustrates that the adoption of SEO has a strong relationship with digital capabilities in SMEs. Companies that have better digital capacities can better develop advanced SEO strategies and incorporate them into the general strategies of the business. This observation is consistent with the resource-based perspective, which proposes that firm-specific capabilities are very essential in attaining sustainable competitive advantage (Khin and Ho, 2019).

Third, the paper presents empirical data regarding the use of the Technology Acceptance Model (TAM) when dealing with the use of the model about digital marketing adoption among SMEs. Although TAM has been largely used in information systems studies, its application in explaining the adoption of SEO in the digital marketing strategies by SMEs is minimal. The findings indicate that the perceived usefulness and perceived ease of use are strong predictors of adopting SEO practices and therefore present TAM to new research setting to be useful (Alalwan, 2020).

Lastly, this study hypothesizes and empirically supports a synthesized conceptualized model, which integrates technology adoption theory, digital capability development, and perspectives on competitive advantage. This combined method adds to theoretical literature because it proves that the adoption of technologies is not enough to enhance the performance of firms. Rather, competitive advantage appears in the case of the adoption of the technology which is complemented by organizational capability and strategic alignment.

6.3 Managerial Implications

The results of this research have several implications to the SME managers and practitioners in digital marketing.

To begin with, SMEs are advised to consider SEO as a strategic investment and not a short-term marketing strategy. The repetitive process of creating content, optimizing the websites, and monitoring the performance are important to sustainable search engine rankings. Companies that

use the strategic approach to SEO have greater chances of enjoying the long-term rewards, which include visibility, acquisition of customers and credibility of the brand.

Second, the findings reflect the need to acquire digital skills and capabilities in SMEs. The absence of technical expertise is one of the greatest obstacles to successful implementation of SEO. SMEs can overcome this problem by investing in the training of the employees, hiring digital marketing firms, or using easy to use SEO tools (Sun et al., 2022).

Third, search engine optimization must be incorporated into a more comprehensive digital marketing approach and not implemented as a single marketing instrument. The companies that integrate SEO with other digital marketing activities like content marketing, social media marketing, and data-driven decision-making are more likely to perform better in marketing (Scuotto et al., 2021).

Moreover, policymakers and business support organizations can use the findings to create insights. Training, availability of digital infrastructure and financial assistance in SME digitalization can assist the SMEs to overcome their resource constraints and increase their competitiveness in the digital markets. The initiatives could have a significant role in digitalizing SMEs and enhancing economic growth (Eggers et al., 2020).

6.4 Limitations

Although this study has made contribution to the body of research on digital marketing and SME competitiveness, it has several limitations that must be considered when analyzing the results.

To begin with, the research design used in the study was a cross-sectional research design, which captures the data at a particular time. Although this is widely applied in research investigating the adoption of technology and the performance of organizations, it cannot be used to determine the progression of SEO adoption and digital marketing capabilities across time. Digital transformation is a dynamic process whereby; companies do not make immediate new capacities and then change their strategies because of technological changes and market conditions. Thus, longitudinal research design may offer more insightful information regarding the impact of the utilization of SEO strategy on the competitiveness and performance of SMEs in a longer period. Research in the future may follow companies over several years to gain a clearer understanding of the long-term impact of the adopting of digital marketing on organizational performance.

Second, the research was based on the self-reported survey data on the sample of the SME managers and business owners. Even though the approach is popular among research concerning digital marketing and SME management, self-reported responses can cause possible biases. The respondents can also overrate the usefulness of their digital marketing activities or they can make subjective judgments of the performance of their firms. These biases would affect the precision of the reported relationships between the adoption of SEO and firm performance. This limitation can be overcome in future research by using objective measures of performance, including information on the website analytics, digital traffic, or conversion rates, or financial performance to supplement the findings of the survey research.

Third, the study was based on SMEs that have already to some degree embraced digital marketing. Consequently, the results might be not quite applicable to the firms that are at the very beginning of the digital transformation or do not use digital marketing strategies yet. The challenges and barriers may be different to the SMEs that had not implemented digital technologies in comparison with the firms that have already become digital. The future research might thus examine the digital marketing adoption in lesser digitally endowed SMEs to understand more about the impediments to digital transformation.

On balance, although these limitations cannot weaken the validity of the study, they outline the great possibilities of future studies to delve deeper into the relationship between the factors of SEO adoption, digital capacities, and the competitiveness of SMEs in digital marketplaces.

6.5 Future Research

The results of the current study present several prospects to the future research that can further enlarge the knowledge regarding search engine optimization (SEO) adoption and its influence on competitiveness and performance of small and medium-sized enterprises (SMEs). Since the digital technologies are dynamic, the findings of this research can be extended in future research to determine other facets of digital marketing and organizational performance.

To start with, the longitudinal research design can be adopted in future study to analyze the impact of SEO adoption on the SME performance over a long period. The current research was of a cross-sectional study design, which describes relationships at one time point between variables. Nonetheless, digital transformation and uptake of digital marketing are dynamic activities that

change as the firms grow in features and changes in technology. Longitudinal studies were able to follow SMEs through a long period of time and how the adoption of SEO strategies evolve and what effects they have in enhancing the performance and competitiveness of firms in the long term. This kind of research would help learn more about the value of digital marketing capabilities of SMEs over the long-term.

Second, a mixed-methods research design that will understand how to integrate both the quantitative and the qualitative research methods can be applied in future research. Although quantitative surveys give a researcher the opportunity to compare trends among a significant number of firms, qualitative interviews may be used to get deeper insights into the managerial decision-making and organizational processes associated with the use of digital marketing. Interviewing SME managers, digital marketing experts, or industry experts may assist the researchers in better comprehending the strategic reasons, difficulties, and internal organizational factors which may contribute to the implementation of SEO strategies.

Third, a study on cross-country and cross-industry comparisons can be pursued in the future. Implementing and applying the concept of digital marketing can be less effective (or not effective) in the economic landscape, technological facilities, and industries. The comparison of the countries or industries may contribute to the discovery of contextual factors that shape the adoption of SEO and the effectiveness of digital marketing. This kind of research would offer a good source of information about how institutional environments, market conditions, and industry characteristics influence SME digital transformation.

Lastly, further research may also focus on investigating how recent technologies like artificial intelligence (AI), machine learning, and big data analytics may help enhance the performance of SEO strategies and digital marketing. High technologies are being integrated more in the digital marketing platforms and thus allowing firms to analyze the behavior of customers, predict search trend, and make marketing strategies more effective. Further investigation of the role of these technologies in SEO activities and the competitiveness of SMEs might be a valuable research opportunity in the future (Saura et al., 2021).

Altogether, with the ongoing digital technologies development and transformation of the business world, the future research on the possibilities of digital marketing and competitiveness of SMEs will be a topical issue of academic and practical interest.

6.6 Policy Implications

This study has significant implications to policymakers who would like to enable small and medium-sized enterprises (SMEs) to undertake a digital transformation and remain competitive. Since contemporary markets are becoming highly digital, SMEs need to build effective digital marketing capacities to ensure that they tend to be competitive. The findings of this study, however, have shown that numerous SMEs have structural challenges including insufficient financial resources, absence of ready team digital capabilities, and inadequate technological infrastructure. These issues also show that special policy interventions should be conducted under the assumption of SME digitalization and the enhancement of their online access to digital marketing opportunities, e.g., search engine optimization (SEO) (Kraus et al., 2021; Eggers et al., 2020).

The financial and institutional support of SME digital transformation is one of the most crucial implications of the policies. A few SMEs find it difficult to invest in digital technologies because of their limited budgets and doubts on benefits of digital strategy marketing in the long run. Governments can also contribute to this by introducing financial incentives, grants, tax credits, or subsidized programs to lure SMEs to invest in digital marketing infrastructure and training. With the help of such support, smaller companies will be able to use more sophisticated digital marketing tools, enhance their online presence, and become more competitive in the digital markets (Müller et al., 2021).

The other significant policy priority is the creation of the digital skills and training initiatives to SME managers and employees. The research also found that digital knowledge and technical expertise are one of the significant deterrents to the successful execution of SEO strategies. To overcome this problem, policymakers can promote the digital education program, trainings, and collaboration with universities or organizations in the industrial sector to equip SMEs with professional skills and knowledge needed to successfully use digital marketing strategies. Increasing the digital competencies of SMEs can be a factor that will considerably increase the capacity to use SEO and data analytics among other digital tools to increase marketing performance (Sun et al., 2022).

Also, policymakers can encourage SMEs to spread digital awareness and share knowledge-sharing initiatives. The status of many small businesses has been that of doubt concerning the strategic advantages of digital marketing and the possible payback that comes along with the SEO strategies.

Such social programs as digital innovation hubs, advisory services, and mentorship programs can assist SMEs to achieve access to best practices, digital transformation practice, and design effective digital marketing strategies (Maroufkhani et al., 2022).

Lastly, the governments must be aimed at enhancing digital infrastructure and access to technology. They are good internet connection, access to online platforms, and availability of inexpensive technological solutions, which ensure that SMEs can engage in digital markets. Digital infrastructure and national digital strategies can also play a significant role in ensuring that SMEs are able to embrace the use of digital marketing technologies and compete internationally (Vial, 2019).

7. References

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